

Ecuador — Ley Orgánica de Protección de Datos Personales

Discovery and masking of Ecuadorian personal data: cédula and RUC with a valid check digit, quasi-identifiers, the sensitive data of article 4 — which names the cultural identity, the migratory status and the judicial record no neighbouring law does —, health data, credit data and criminal proceedings.

82

DOMAINS

141

CLASSIFIERS

112

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About this profile set

This profile set finds the personal data in an Ecuadorian database and masks it, so that a non-production copy — for testing, development, training or analysis — no longer identifies anyone.

Masking names and cédulas alone is not enough: a birth date, a sex and a small parroquia already point at one person. The set works in the layers where the risk is:

- **Direct identifiers** — cédula and RUC, foreigner documents, passport, civil registry, IESS and pension numbers, student code, professional registration, names, contact, address, accounts, cards, devices, cookies, plates, property, free text.
- **Quasi-identifiers** — birth date, age, sex, parroquia and cantón with their DPA code, postal code, ciudadelita, marital status, occupation, employer, education.
- **Sensitive data** — those of article 4: etnia, **identidad cultural**, identidad de género, religión, ideología, filiación política, **condición migratoria**, orientación sexual, health, biometric and genetic data, and the data of stateless people and refugees the same article names.
- **Health data** — sensitive by article 4 and a special category by article 25 c): clinical record, insurance, ICD-10, procedures, medicines, HIV, blood group, disability — a special category of its own under article 25 d) —, occupational, reproductive and mental health.
- **Credit data** — the *datos personales crediticios* articles 28 and 29 give a regime of their own, plus debt, income, assets, pensions, social programmes, the Registro Social and housing tenure.
- **Criminal records** — the **pasado judicial** article 4 lists among the sensitive data, and court file numbers.

In a Masking Engine, the set is a profile set with its classifiers, the domains they vote for, the algorithms those domains name — and the algorithms those rely on — and the lookup files they read. The appendices list each of them with its full configuration.

The assignment threshold is 60%. An unambiguous column name reaches it on its own; an ambiguous one — NOMBRE , DOCUMENTO , EXPEDIENTE , CARGO — needs the values to agree. A column with a meaningless name (CAMPO1 , OBS) is still found when its values have a recognizable shape or vocabulary: a cédula, a thirteen-digit RUC, a mobile starting 09, an address with ciudadelita, manzana and villa, a plate, a parroquia.

Column names are recognized in Spanish, in the abbreviations of Ecuadorian systems (NUM_DOC , FEC_NAC , DPA , RUC , IESS , AMIE) and in English.

Before masking, inventory the binary columns (BLOB, images, PDF). Onboarding processes often keep a photo of the cédula — with the fingerprint and the signature — and a validation selfie. Report those columns as residual risk.

The LOPDP in brief

The **Ley Orgánica de Protección de Datos Personales** was published in the Fifth Supplement of Official Register 459 of **26 May 2021**, and its **Reglamento General** in Official Register Supplement 435 of **13 November 2023**. The sanctions regime has been applicable since 26 May 2023. The law is the closest in the region to the European model, and it takes its definitions from it.

What matters for a copy of production data:

- **Personal data** is the datum that identifies or makes identifiable a natural person, directly or indirectly (art. 4). The **titular** is always a natural person: a company is not a data subject.
- **Sensitive data** is the longest list in the region (art. 4): etnia, identidad de género, **identidad cultural**, religión, ideología, filiación política, **pasado judicial**, **condición migratoria**, orientación sexual, salud, datos biométricos, datos genéticos, and the data of stateless people and refugees who need international protection — closing with an open clause for anything whose improper processing may give rise to discrimination.
- **Special categories** (art. 25) are four: sensitive data, the data of **children and adolescents**, health data, and the data of **persons with disabilities and of their substitutes**. Article 26 forbids processing sensitive data unless one of the grounds it lists applies.
- **Anonymization** is the application of measures that prevent the identification or **re-identification** of a natural person without disproportionate effort (art. 4). **Pseudonymization** is processing after which the data can no longer be attributed to a subject without using additional information, provided that information is kept apart and under technical and organizational measures (art. 4) — the European pair, word for word.
- **Where they appear.** The law does **not apply** to anonymized data, so long as the subject cannot be identified; as soon as the data stops being dissociated or anonymous, its processing falls back under the law (art. 2 c). Article 37 lists **anonymization, pseudonymization or encryption** first among the security measures. Article 31.2 asks that health data be **previously anonymized or pseudonymized whenever possible**.
- **Security.** A breach is notified to the Authority and to ARCOTEL as soon as possible and at the latest within **five days**; a processor notifies the controller within two (art. 43). Articles 39 to 42 ask for protection by design and by default, risk analysis and an impact assessment.

Credit data have a regime of their own. Article 4 defines the *datos personales crediticios* as those that make up the economic behaviour of a natural person; article 28 allows processing them to analyse solvency and **forbids any secondary purpose**, and forbids communicating them once five years have passed since the obligation became due.

Anonymization, pseudonymization and non-production copies

Ecuador writes the two concepts the way the European regulation does. **Anonymization** prevents identification **and re-identification** without disproportionate effort, and takes the data out of the law (art. 2 c). **Pseudonymization** accepts that the data can be attributed to its subject again *with additional information*, and for that reason requires that information to be kept apart and under technical and organizational measures.

The algorithms of this set are deterministic for the Masking Engine's key: the same cédula masks the same way in every table, which keeps joins working. That key is exactly the *additional information* article 4 speaks of. The masked copy is **pseudonymized**, not anonymized, and remains personal data for whoever holds the key and production.

- Keep the engine key apart from the copy and under technical and organizational measures — that is what article 4 asks for pseudonymization to count, and article 37 counts it as a security measure.
- For health data, article 31.2 asks for anonymization or pseudonymization *whenever possible*: a masked copy of a clinical database is the ordinary way to comply with it.
- If what you need is anonymized data — out of the law by article 2 c) —, measure the re-identification risk of the quasi-identifiers that remain; the standard of article 4 is higher than this set meets.

How sensitive data is discovered

A profiling job looks at every column of its rule set and asks each classifier of the profile set how confident it is that the column belongs to its domain. The set uses the four classifier frameworks of the Masking Engine:

- **Column name** (Path) — regular expressions on the column name. Each rule has a match strength: the confidence it gives when it matches.
- **Data type** (Type) — accepts the column type and length, or rules the domain out. It never assigns a domain on its own.
- **Value format** (Regex) — tests a sample of the values against patterns, some with a check digit. The confidence follows the share of values that match.
- **Known values** (List) — looks the sampled values up in lists; for names, word by word.

Evidence from the name and from the values is combined per domain, and the column is assigned to the domain with the highest confidence when it reaches the assignment threshold of the profile set — **60%** here. Values that do not match a format or a list count against their domain by the classifier's reject strength.

Column-name rules are shown as their alternatives. They ignore case and must not be glued to other letters:

`rut` finds `RUT_CLIENTE` and `CLIENTE_RUT` but not `RUTA`. In them, `_?` is an optional underscore, `(a|b)` either form, `s?` an optional letter and `[a-z_]*` any continuation.

Profiling proposes; it does not decide. Review the domains assigned in the inventory before running a masking job, and look for columns the set left unassigned.

How data is masked

Each domain names one algorithm, which a masking job applies to every column the inventory assigns to the domain. Algorithms can be composed: a **Regex Decompose** splits a value into parts and gives each part its own action; a **String Algorithm Chain** applies algorithms one after another. The tree under each domain shows everything its algorithm uses; Appendix A has every configuration in full.

The algorithms are deterministic for their key: the same value masks the same way in every table, database and job, so joins between masked systems keep working — as long as every engine uses the same keys.

The masked examples in this document were produced with one particular key. On an engine with other keys the masked values differ; their shape does not.

Essential and extended packs

The set can be adopted whole or in part. The **extended pack** is everything this document describes. The **essential pack** is the minimum for the law — identity documents, names, contact details, address and birth date. To adopt only the essential pack, create on the engine just its domains, the classifiers that vote for them and the algorithms and files those use, and put only those classifiers in the profile set.

PACK	DOMAINS	CLASSIFIERS	ALGORITHMS	FILES
Essential	12	25	17	5
Extended	82	141	112	80

The essential pack holds these domains, marked **Essential** in the tables and chapters:

- Cédula de identidad — EC_L1_CEDULA
- RUC — EC_L1_RUC
- Foreigner documents — EC_L1_DOCUMENTO_EXTRANJERO
- Passport — EC_L1_PASAPORTE
- Given name — EC_L1_NOMBRE
- Surname — EC_L1_APELLIDO
- Full name — EC_L1_NOMBRE_COMPLETO
- E-mail — EC_L1_EMAIL
- Phone — EC_L1_TELEFONO
- Address — EC_L1_DIRECCION
- Address complement — EC_L1_DIRECCION_COMPLEMENTO
- Birth date — EC_L2_FECHA_NACIMIENTO

With fewer domains competing, a column that a domain of the extended pack would claim by its name can be proposed for an essential domain instead — a numeric code, for instance, can look like a telephone number. Review the profiling result.

Domains at a glance

Every domain of the set, how it is found and the algorithm that masks it.

DOMAIN		FOUND BY		ALGORITHM
DIRECT IDENTIFIERS				
Cédula de identidad ESSENTIAL		COLUMN NAME	DATA TYPE	VALUE FORMAT
EC_L1_CEDULA				EC_DOCUMENTO
RUC ESSENTIAL		COLUMN NAME	VALUE FORMAT	
EC_L1_RUC				EC_RUC
Foreigner documents ESSENTIAL		COLUMN NAME		
EC_L1_DOCUMENTO_EXTRANJERO				EC_CM_ALFANUM
Passport ESSENTIAL		COLUMN NAME	VALUE FORMAT	
EC_L1_PASAPORTE				EC_CM_ALFANUM
Civil registry records		COLUMN NAME		
EC_L1_ACTA				EC_CM_ALFANUM
IESS, ISSFA and ISSPOL		COLUMN NAME		
EC_L1_SEGURO_SOCIAL				EC_CM_ALFANUM
Student code		COLUMN NAME		
EC_L1_CODIGO_ESTUDIANTE				EC_CM_ALFANUM
Professional registration		COLUMN NAME		
EC_L1_REGISTRO_PROFESIONAL				EC_CM_ALFANUM
Given name ESSENTIAL		COLUMN NAME	KNOWN VALUES	
EC_L1_NOMBRE				EC_NOMBRES
Surname ESSENTIAL		COLUMN NAME	KNOWN VALUES	
EC_L1_APELLIDO				EC_APELLIDOS
Full name ESSENTIAL		COLUMN NAME	KNOWN VALUES	
EC_L1_NOMBRE_COMPLETO				EC_NOMBRE_COMPLETO
Business name		COLUMN NAME		
EC_L1_RAZON_SOCIAL				EC_RAZON_SOCIAL
E-mail ESSENTIAL		COLUMN NAME	DATA TYPE	VALUE FORMAT
EC_L1_EMAIL				EC_EMAIL
Phone ESSENTIAL		COLUMN NAME	VALUE FORMAT	
EC_L1_TELEFONO				EC_TELEFONO
Address ESSENTIAL		COLUMN NAME	DATA TYPE	VALUE FORMAT
EC_L1_DIRECCION				EC_DIRECCION
Address complement ESSENTIAL		COLUMN NAME		
EC_L1_DIRECCION_COMPLEMENTO				EC_CM_ALFANUM
Bank account		COLUMN NAME		
EC_L1_CUENTA_BANCARIA				EC_CM_DIGITOS
Payment card		COLUMN NAME	VALUE FORMAT	
EC_L1_TARJETA				EC_TARJETA
Digital wallet		COLUMN NAME	VALUE FORMAT	
EC_L1_BILLETERA				EC_CM_ALFANUM

DOMAIN	FOUND BY			ALGORITHM
IP address EC_L1_IP	COLUMN NAME	VALUE FORMAT		EC_IP
Device identifier EC_L1_DISPOSITIVO	COLUMN NAME	VALUE FORMAT		EC_CM_HEX
Cookies and tracking identifiers EC_L1_COOKIE	COLUMN NAME			EC_CM_ALFANUM
Vehicle plate EC_L1_PLACA	COLUMN NAME	VALUE FORMAT		EC_CM_DIGITOS_LETRAS
Vehicle and licence EC_L1_VEHICULO	COLUMN NAME	VALUE FORMAT		EC_CM_ALFANUM
Property EC_L1_INMUEBLE	COLUMN NAME			EC_CM_ALFANUM
Customer and contract numbers EC_L1_CONTRATO	COLUMN NAME			EC_CM_ALFANUM
User and social media EC_L1_USUARIO	COLUMN NAME			EC_CM_ALFANUM
Credentials EC_L1_CREDENCIAL	COLUMN NAME			EC_REDACTAR
Geolocation EC_L1_GEOLOCALIZACION	COLUMN NAME	VALUE FORMAT		EC_COORDENADA
Free text EC_L1_TEXTO_LIBRE	COLUMN NAME	DATA TYPE	VALUE FORMAT	EC_TEXTO_LIBRE
QUASI-IDENTIFIERS				
Birth date ESSENTIAL EC_L2_FECHA_NACIMIENTO	COLUMN NAME	DATA TYPE		EC_FECHA_NACIMIENTO
Birth year EC_L2_ANIO_NACIMIENTO	COLUMN NAME	DATA TYPE		EC_ANIO
Age EC_L2_EDAD	COLUMN NAME	DATA TYPE		EC_EDAD
Personal event dates EC_L2_FECHA_EVENTO	COLUMN NAME	DATA TYPE		EC_FECHA_EVENTO
Sex EC_L2_SEXO	COLUMN NAME	KNOWN VALUES		EC_SEXO
Parroquia and cantón EC_L2_PARROQUIA	COLUMN NAME	KNOWN VALUES		EC_PARROQUIA
DPA code EC_L2_DPA	COLUMN NAME	VALUE FORMAT		EC_DPA
Ciudadela, urbanización and barrio EC_L2_CIUDADELA	COLUMN NAME			EC_CIUDADELA
Postal code EC_L2_CODIGO_POSTAL	COLUMN NAME	DATA TYPE		EC_CODIGO_POSTAL
Marital status EC_L2_ESTADO_CIVIL	COLUMN NAME	KNOWN VALUES		EC_ESTADO_CIVIL
Occupation EC_L2_OCUPACION	COLUMN NAME			EC_OCUPACION_O_CODIGO

DOMAIN	FOUND BY	ALGORITHM
Employer EC_L2_EMPLEADOR	COLUMN NAME	EC_EMPLEADOR
Education EC_L2_NIVEL_INSTRUCCION	COLUMN NAME KNOWN VALUES	EC_NIVEL_INSTRUCCION
School EC_L2_INSTITUCION_EDUCATIVA	COLUMN NAME	EC_INSTITUCION_EDUCATIVA
Dependants and children EC_L2_PERSONAS_A_CARGO	COLUMN NAME DATA TYPE	EC_CONTEO_LIMITADO
SENSITIVE DATA (ART. 4)		
Ethnicity EC_L3_ETNIA	COLUMN NAME KNOWN VALUES	EC_AUTOIDENTIFICACION
Cultural identity EC_L3_IDENTIDAD_CULTURAL	COLUMN NAME KNOWN VALUES	EC_IDENTIDAD_CULTURAL
Language EC_L3_LENGUA	COLUMN NAME KNOWN VALUES	EC_LENGUA
Migratory status EC_L3_CONDICION_MIGRATORIA	COLUMN NAME KNOWN VALUES	EC_CONDICION_MIGRATORIA
Nationality EC_L3_NACIONALIDAD	COLUMN NAME KNOWN VALUES	EC_NACIONALIDAD_PAIS
Religion EC_L3_RELIGION	COLUMN NAME KNOWN VALUES	EC_RELIGION
Ideology EC_L3_IDEOLOGIA	COLUMN NAME KNOWN VALUES	EC_IDEOLOGIA
Political affiliation EC_L3_FILIACION_POLITICA	COLUMN NAME KNOWN VALUES	EC_PARTIDO
Trade union membership EC_L3_AFILIACION_SINDICAL	COLUMN NAME KNOWN VALUES	EC_CATEGORIA_SUPRIMIDA
Biometric data EC_L3_BIOMETRICO	COLUMN NAME DATA TYPE	EC_REDACTAR
Genetic data EC_L3_GENETICO	COLUMN NAME	EC_REDACTAR
Sexual orientation EC_L3_ORIENTACION_SEXUAL	COLUMN NAME KNOWN VALUES	EC_ORIENTACION_SEXUAL
Gender identity EC_L3_IDENTIDAD_GENERO	COLUMN NAME KNOWN VALUES	EC_IDENTIDAD_GENERO
Victims of violence EC_L3_VICTIMA	COLUMN NAME	EC_CATEGORIA_SUPRIMIDA
HEALTH DATA		
Health identifiers EC_SALUD_IDENTIFICADOR	COLUMN NAME	EC_CM_ALFANUM
Health insurance EC_SALUD_COBERTURA	COLUMN NAME KNOWN VALUES	EC_COBERTURA_SALUD
Diagnosis and ICD-10 EC_SALUD_DIAGNOSTICO	COLUMN NAME VALUE FORMAT	EC_DIAGNOSTICO

DOMAIN	FOUND BY	ALGORITHM
Procedures EC_SALUD_PROCEDIMIENTO	COLUMN NAME	EC_PROCEDIMIENTO
Medicines EC_SALUD_MEDICAMENTO	COLUMN NAME KNOWN VALUES	EC_MEDICAMENTO
Clinical text EC_SALUD_TEXTO_CLINICO	COLUMN NAME DATA TYPE	EC_TEXTO_LIBRE
Test results EC_SALUD_RESULTADO_EXAMEN	COLUMN NAME	EC_CM_ALFANUM
HIV and sexually transmitted infections EC_SALUD_VIH	COLUMN NAME	EC_CATEGORIA_SUPRIMIDA
Blood group and Rh EC_SALUD_GRUPO_SANGUINEO	COLUMN NAME KNOWN VALUES	EC_GRUPO_SANGUINEO
Disability EC_SALUD_DISCAPACIDAD	COLUMN NAME KNOWN VALUES	EC_DISCAPACIDAD
Occupational health EC_SALUD_OCUPACIONAL	COLUMN NAME KNOWN VALUES	EC_APTITUD_LABORAL
Sexual and reproductive health EC_SALUD_REPRODUCTIVA	COLUMN NAME	EC_CATEGORIA_SUPRIMIDA
Mental health EC_SALUD_MENTAL	COLUMN NAME	EC_CATEGORIA_SUPRIMIDA
CREDIT AND SOCIOECONOMIC DATA		
Credit history EC_FIN_HISTORIAL_CREDITICIO	COLUMN NAME	EC_HISTORIAL_CREDITICIO
Debt EC_FIN_DEUDA	COLUMN NAME DATA TYPE	EC_VALOR_DEUDA
Income and pay EC_FIN_INGRESOS	COLUMN NAME DATA TYPE	EC_INGRESOS
Assets EC_FIN_PATRIMONIO	COLUMN NAME DATA TYPE	EC_PATRIMONIO
Pension and benefits EC_FIN_PENSION	COLUMN NAME DATA TYPE	EC_VALOR_PENSION
Social programmes EC_FIN_PROGRAMA_SOCIAL	COLUMN NAME KNOWN VALUES	EC_PROGRAMA_SOCIAL
Registro Social EC_FIN_REGISTRO_SOCIAL	COLUMN NAME KNOWN VALUES	EC_REGISTRO_SOCIAL
Housing tenure EC_FIN_VIVIENDA	COLUMN NAME KNOWN VALUES	EC_TENENCIA_VIVIENDA
JUDICIAL RECORD		
Judicial record EC_PENAL_ANTECEDENTES	COLUMN NAME	EC_ANTECEDENTES
Court file EC_PENAL_EXPEDIENTE	COLUMN NAME VALUE FORMAT	EC_EXPEDIENTE_JUDICIAL

Direct identifiers

Data that identifies a person on its own. The cédula is the key of almost every Ecuadorian record, and a natural person's RUC is that same cédula followed by 001. Both come out valid, because applications validate them, and both mask the same way everywhere. A RUC that belongs to a company is left as it is: article 4 makes only a natural person a data subject.

8.1 Cédula de identidad

EC_L1_CEDULA

ESSENTIAL COLUMN NAME DATA TYPE VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The ten-digit number of the Registro Civil, 1710034065, the key of almost every Ecuadorian record. The first two digits are the province of issue — 01 to 24, or 30 for a citizen registered abroad —, the third says the holder is a natural person, and the tenth is a check digit.

The same domain takes RUCs: a document column often holds both.

HOW IT IS FOUND

COLUMN NAME

EC_L1_CEDULA - Path

90%

c[eé]dula(_?(de_)?(identidad|ciudadan[ií]a))?

ci

n(o|ro|um|umero)?

(c[eé]dula|documento|doc|identificaci[oó]n|id(ent)?)

doc_?identidad

nro_?doc

num_?doc

identificacion

(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|trabajador|colaborador|alumno|estudiante|deudor|codeudor|garante|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elector|votante|contribuyente|proveedor|socio|conyuge|conviviente|padre|madre|hijo|familiar|jubilado|pensionista|aportante|derechohabiente)?(cedula|doc|documento)

cedula_?

(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|trabajador|colaborador|alumno|estudiante|deudor|codeudor|garante|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elector|votante|contribuyente|proveedor|socio|conyuge|conviviente|padre|madre|hijo|familiar|jubilado|pensionista|aportante|derechohabiente)

60%

documento

doc

DATA TYPE EC_L1_CEDULA - Type

Accepts text (length 10 or more) or number (length 10 or more); any other type rules the domain out.

VALUE FORMAT EC_L1_CEDULA - Regex

PATTERN	STRENGTH	NOTES
(0[1-9] 1\d 2[0-4] 30)[0-5]\d{7}		80%
\d{10}		50%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Another valid cédula, with the province of issue and the natural-person digit kept and the check digit recomputed. A value shaped like a RUC is masked as a RUC.

EC_DOCUMENTO REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: EC_CM_ALFANUM .

EC_RUC REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_CM_DIGITOS .

EC_CEDULA_LUHN PAYMENT CARD

Replaces the card number with another that passes the Luhn check.

EC_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EC_CEDULA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_CM_DIGITOS .

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
1710034065	→	1718687906
1710034065001	→	1718687906001
1790010937001	→	1790010937001

8.2

RUC

EC_L1_RUC

ESSENTIAL

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

The thirteen-digit taxpayer number of the SRI. For a natural person it is the cédula followed by the establishment code 001 ; the third digit decides — 0 to 5 a natural person, 6 a public body, 9 a private company.

HOW IT IS FOUND

COLUMN NAME EC_L1_RUC - Path

90%

```
ruc[a-z0-9]* n(o|ro|um|umero)?ruc registro?[uú]nico?(de_)?contribuyentes?
(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|
trabajador|colaborador|alumno|estudiante|deudor|codeudor|garante|solicitante|propietario|arrend
datario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elector|vota
nte|contribuyente|proveedor|socio|conyuge|conviviente|padre|madre|hijo|familiar|jubilado|pensi
onista|aportante|derechohabiente)?ruc ruc_?
(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|t
rabajador|colaborador|alumno|estudiante|deudor|codeudor|garante|solicitante|propietario|arrend
atario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elector|vota
nte|contribuyente|proveedor|socio|conyuge|conviviente|padre|madre|hijo|familiar|jubilado|pensi
onista|aportante|derechohabiente) id_?tributario tax_?id
```

VALUE FORMAT EC_L1_RUC - Regex

PATTERN	STRENGTH	NOTES
(0[1-9] 1\d 2[0-4] 30)[0-9]\d{7}00\d		85%
\d{13}		50%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

A natural person's RUC is masked as the cédula it carries, so both agree. A RUC whose third digit is 6 or 9 belongs to a legal person, which article 4 does not make a data subject: it is left exactly as it is.

EC_RUC REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_CM_DIGITOS .

EC_CEDULA_LUHN PAYMENT CARD

Replaces the card number with another that passes the Luhn check.

EC_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
1710034065001	→	1718687906001
1790010937001	→	1790010937001
1760013210001	→	1760013210001

8.3 Foreigner documents

EC_L1_DOCUMENTO_EXTRANJERO

ESSENTIAL COLUMN NAME CHARACTER MAPPING

WHAT IT IS

The cédula de extranjería, the refugee or applicant card, the visa and the migration procedure numbers. The **condición migratoria** these carry is sensitive data by article 4 and has a domain of its own.

HOW IT IS FOUND

COLUMN NAME EC_L1_DOCUMENTO_EXTRANJERO - Path

85% c[eé]dula_?(de_)?extranjer[ií]a carn[eé]_?(de_)?(extranjer[ií]a|refugiado|solicitante) documento_?(de_)?extranjer[oa] doc_?extranjero visa(_? (n(o|ro|um|umero)|tipo|categor[ií]a)? n(o|ro|um|umero)_?(visa|refugiado|extranjer[ií]a) permiso_?(de_)?(residencia|trabajo) amparo_?(legal|migratorio) n(o|ro|um|umero)_?(de_)? (tr[aá]mite_?migratorio|movimiento_?migratorio)

HOW IT IS MASKED

Character by character, keeping length and format.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
R123456	→	Q801473

8.4 Passport

EC_L1_PASAPORTE

ESSENTIAL COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

Ecuadorian passport numbers and foreign ones.

HOW IT IS FOUND

COLUMN NAME EC_L1_PASAPORTE – Path

90% `pasaporte[a-z0-9]*` `n(o|ro|um|umero)?pasaporte` `passport[a-z0-9]*`

VALUE FORMAT EC_L1_PASAPORTE – Regex

PATTERN	STRENGTH	NOTES
<code>[A-Z]{1,3}\d{6,8}</code>		40%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Character by character, keeping length and format.

EC_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
A1234567	→ U2438404

8.5

Civil registry records

EC_L1_ACTA

COLUMN NAME **CHARACTER MAPPING**

WHAT IT IS

The acta, tomo and folio of the birth, marriage, union of fact and death records the Dirección General de Registro Civil, Identificación y Cédulación keeps.

HOW IT IS FOUND

COLUMN NAME EC_L1_ACTA – Path

85% `acta_(de_)?(nacimiento|matrimonio|defunci[oó]n|uni[oó]n?de?hecho)(_?`
`(n(o|ro|um|umero)|folio|tomo))?` `n(o|ro|um|umero)?acta` `partida_(de_)?`
`(nacimiento|matrimonio|defunci[oó]n)` `tomo(_?(acta|registro))?` `folio(_?(acta|registro))?`
`n(o|ro|um|umero)?(de_)?inscripci[oó]n?civil` `registro?civil`

HOW IT IS MASKED

Character by character, keeping length and format.

EC_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
123/1985	→ 091/7273

COLUMN NAME	CHARACTER MAPPING
-------------	-------------------

WHAT IT IS

The IESS affiliation and employer numbers, the labour history, and the numbers of the armed forces and police funds.

HOW IT IS FOUND

COLUMN NAME	EC_L1_SEGURO_SOCIAL - Path
85%	<code>iess[a-z0-9]*</code> <code>n(o ro um umero)?(iess afiliaci[oó]n afiliado asegurado patronal)</code> <code>c[oó]digo?(de_)?(afiliado asegurado patronal)</code> <code>issfa</code> <code>isspol</code> <code>seguro?social?campesino</code> <code>n(o ro um umero)?(de_)?(historia?laboral aportaciones)</code> <code>biess</code>

HOW IT IS MASKED

Character by character, keeping length and format.

EC_CM_ALFANUM	CHARACTER MAPPING
---------------	-------------------

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
123456789	→ 110858568

COLUMN NAME	CHARACTER MAPPING
-------------	-------------------

WHAT IT IS

The student code, the enrolment number and the university card.

HOW IT IS FOUND

COLUMN NAME	EC_L1_CODIGO_ESTUDIANTE - Path
80%	<code>c[oó]digo?(de_)?(estudiante alumno matr[ií]cula educando)</code> <code>cod?(alumno estudiante)</code> <code>n(o ro um umero)?(de_)?(matr[ií]cula carn[eé]?estudiantil)</code> <code>carn[eé]?(de_)?</code> <code>(estudiante universitario)</code> <code>id?(alumno estudiante)</code> <code>c[oó]digo?amie?(del_)?estudiante</code>

HOW IT IS MASKED

Character by character, keeping length and format.

EC_CM_ALFANUM	CHARACTER MAPPING
---------------	-------------------

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
202012345	→ 239159952

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

The SENESCYT registration number of a degree, published together with the name, and the professional licences.

HOW IT IS FOUND

COLUMN NAME EC_L1_REGISTRO_PROFESIONAL - Path

85% registro?(senescyt|profesional|de?t[ií]tulo) n(o|ro|um|umero)?(de_)?(registro? senescyt|t[ií]tulo|colegiatura) senescyt c[ó]digo?(de_)?t[ií]tulo matr[ií]cula? profesional licencia?profesional

HOW IT IS MASKED

Character by character, keeping length and format.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
1005-12-1234567	→ 0014-51-8129032

ESSENTIAL COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

The given name, one or two: Narcisa, Luis Alberto, María de los Ángeles.

HOW IT IS FOUND

COLUMN NAME EC_L1_NOMBRE - Path

85% primer?nombre segundo?nombre nombres?(de_)?pila pri(mer)?_?nom seg(undo)?_?nom nombre_?1 nombre_?2 first?names? given?names? fname middle?names? nombres (except when followed by completos?, y?apellidos?, apellidos?)

50% nombre (except when followed by list E1) nom (except when followed by list E2) name (except when followed by list E3)

KNOWN VALUES EC_L1_NOMBRE - List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-nombres.txt	395	70%	José, Jose, María, Maria, Juan, Carlos

Values are split into words, and each word is looked up.
A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Word by word, replacing each name with one from the list. de, del and la stay in place. More than four words are masked character by character.

EC_NOMBRES REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: EC_CM_ALFANUM.

EC_PALABRA_NOMBRE REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: EC_NOMBRE .

EC_NOMBRE NAME

Replaces the name with one of the 338 names in ec-nombres.txt .

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
María de los Ángeles	→	Washington de los Patricia
LUIS ALBERTO	→	SISA ELENA

8.10 Surname

EC_L1_APELLIDO

ESSENTIAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The surnames — the father's and the mother's —, in separate columns or together: Zambrano Cedeño . The set carries the Ecuadorian ones, from Zambrano and Loor on the coast to Guamán and Quishpe in the sierra.

HOW IT IS FOUND

COLUMN NAME EC_L1_APELLIDO - Path

85% apellidos? primer_?apellido segundo_?apellido apellido_?(1|2|paterno|materno|casada) ape_?(1|2|pat|mat) ap_?(paterno|materno) last_?names? surnames? family_?names? lname

KNOWN VALUES EC_L1_APELLIDO - List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-apellidos.txt	454	70%	Zambrano, Vera, García, Garcia, Mero, Loor

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Word by word, replacing each surname with one from the list; particles stay in place.

EC_APELLIDOS REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: EC_CM_ALFANUM .

EC_PALABRA_APELLIDO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: EC_APELLIDO .

EC_APELLIDO NAME

Replaces the name with one of the 366 names in ec-apellidos.txt .

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
Zambrano Cedeño	→	Morales Calle

8.11 Full name

EC_L1_NOMBRE_COMPLETO

ESSENTIAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Given names and surnames in one column: Luis Alberto Zambrano Cedeño . Also those of the mother, father, guardian, spouse, partner, emergency contact and reference.

HOW IT IS FOUND

COLUMN NAME

EC_L1_NOMBRE_COMPLETO - Path

90%

```
nombres?_?(completos?|y_?apellidos?|apellidos?)  apellidos?_?(y_?)?nombres?  nombre_?(del_?|de_?la_?)?

(c|cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|trabajador|colaborador|alumno|estudiante|deudor|codeudor|garante|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|contribuyente|madre|padre|conyuge|conviviente|contacto|referencia|victima|denunciante|procesado|testigo|medico|docente|socio|heredero|donante|tutor|jubilado|pensionista|derechohabiente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner)

(c|cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|trabajador|colaborador|alumno|estudiante|deudor|codeudor|garante|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|contribuyente|madre|padre|conyuge|conviviente|contacto|referencia|victima|denunciante|procesado|testigo|medico|docente|socio|heredero|donante|tutor|jubilado|pensionista|derechohabiente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner)_?(nombre|nom|nom_?

(c|cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|trabajador|colaborador|alumno|estudiante|deudor|codeudor|garante|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|contribuyente|madre|padre|conyuge|conviviente|contacto|referencia|victima|denunciante|procesado|testigo|medico|docente|socio|heredero|donante|tutor|jubilado|pensionista|derechohabiente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner)  nombre_?

(madre|padre|tutor|conyuge|conviviente|contacto_?emergencia|referencia)  full_?name  person_?name
```

60%

```
madre  padre  tutor  conyuge  conviviente  representante_?legal  apoderado  garante
beneficiario  heredero  titular  referencia_?(personal|familiar)
```

50%

```
nombre (except when followed by list E1)  nom (except when followed by list E2)  name (except when followed by list E3)
```

KNOWN VALUES

EC_L1_NOMBRE_COMPLETO - List

LIST

VALUES

STRENGTH

FIRST VALUES

ec-detectar-nombres.txt

395

65%

José, Jose, María, Maria, Juan, Carlos

ec-detectar-apellidos.txt

454

65%

Zambrano, Vera, García, Garcia, Mero, Loor

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

In Ecuadorian order: two words are a given name and a surname; three, a given name and two surnames; four, two and two. APELLIDOS, NOMBRES with a comma is masked keeping the comma.

EC_NOMBRE_COMPLETO **REGEX DECOMPOSE**

Tries 7 patterns in order and gives each part of the value its own action; when none matches: EC_CM_ALFANUM .

EC_APELLIDOS **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: EC_CM_ALFANUM .

EC_PALABRA_APELLIDO **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: EC_APELLIDO .

EC_APELLIDO **NAME**

Replaces the name with one of the 366 names in ec-apellidos.txt .

EC_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EC_NOMBRES **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: EC_CM_ALFANUM .

EC_PALABRA_NOMBRE **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: EC_NOMBRE .

EC_NOMBRE **NAME**

Replaces the name with one of the 338 names in ec-nombres.txt .

EXAMPLES

VALUE		MASKED
Luis Alberto Zambrano Cedeño	→	Sisa Elena Morales Calle
Narcisa Zambrano	→	Soledad Morales
ZAMBRANO CEDEÑO, LUIS ALBERTO	→	MORALES CALLE, SISA ELENA

8.12 Business name

EC_L1_RAZON_SOCIAL

COLUMN NAME

SECURE LOOKUP

WHAT IT IS

The business name a natural person trades under, beside the RUC that names her. The company itself is not a data subject; the person behind it is.

HOW IT IS FOUND

COLUMN NAME EC_L1_RAZON_SOCIAL – Path

85%

raz[oó]n?social razonsocial nombre?(comercial|de?la?empresa|empresa)

denominaci[oó]n?social nombre?legal business?name company?name legal?name

HOW IT IS MASKED

Replaced by a fictitious business name.

EC_RAZON_SOCIAL **SECURE LOOKUP**

Replaces the value with one of the 20 values in ec-razones-sociales.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Corporación Comercial Ecuatoriana S.A.	→	Transportes Mitad del Mundo S.A.

8.13 E-mail

EC_L1_EMAIL

ESSENTIAL **COLUMN NAME** **DATA TYPE** **VALUE FORMAT** **REGEX DECOMPOSE**

WHAT IT IS

Personal and corporate e-mail addresses.

HOW IT IS FOUND

COLUMN NAME EC_L1_EMAIL – Path

85% e?mail[a-z0-9]* mail[a-z0-9]* correo(?:electr[oó]nico)?[a-z0-9]* dir?correo
direcci[oó]n?(de_)?correo direcci[oó]n?electr[oó]nica

DATA TYPE EC_L1_EMAIL – Type

Accepts text (length 5 or more); any other type rules the domain out.

VALUE FORMAT EC_L1_EMAIL – Regex

PATTERN	STRENGTH	NOTES
[A-Za-z0-9._%+'\-]+@[A-Za-z0-9.\-]+\.[A-Za-z]{2,24}	100%	

A value that matches none counts 20% against the domain.

HOW IT IS MASKED

User and domain are mapped character by character, and the top-level domain — .com , .com.ec — becomes .test , reserved for tests: a masked address can never deliver.

EC_EMAIL **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: EC_CM_ALFANUM .

EC_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
narcisa.zambrano@gmail.com	→ dipcola.gojmyeju@pgoeb.test

8.14 Phone

EC_L1_TELEFONO

ESSENTIAL **COLUMN NAME** **VALUE FORMAT** **REGEX DECOMPOSE**

WHAT IT IS

Mobiles of ten digits starting 09 , and landlines with a one-digit area code after the zero — 02 Pichincha, 04 Guayas, 07 Azuay and El Oro — and seven digits.

HOW IT IS FOUND

COLUMN NAME EC_L1_TELEFONO – Path

85% tel tel[eé]fono[a-z0-9]* tel?
(fijo|casa|oficina|trabajo|contacto|celular|m[oó]vil|convencional) celular[a-z0-9]* cel
m[oó]vil whats?app convencional n(o|ro|um|umero)?
(tel|tel[eé]fono|celular|cel|contacto) fax phone[a-z0-9]* mobile[a-z0-9]*

VALUE FORMAT EC_L1_TELEFONO – Regex

PATTERN	STRENGTH	NOTES
(\+?593[\s-]?)?0?9\d{2}[\s-]?\d{3}[\s-]?\d{3}	90%	
\(0?[2-7]\)?[\s-]?\d{3}[\s-]?\d{4}	50%	

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

+593 , the 9 of a mobile and the area code of a landline stay; the rest of the digits are mapped and the spacing of the original is kept, so the number still says mobile or landline and which region.

EC_TELEFONO **REGEX DECOMPOSE**

Tries 3 patterns in order and gives each part of the value its own action; when none matches: EC_CM_DIGITOS .

EC_CM_DIGITOS **CHARACTER MAPPING**

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
+593 99 123 4567	→	+593 93 743 6352
0991234567	→	0937436352
02 234-5678	→	02 495-1769

8.15 Address

EC_L1_DIRECCION

ESSENTIAL **COLUMN NAME** **DATA TYPE** **VALUE FORMAT** **SECURE LOOKUP**

WHAT IT IS

The address with the main street and the cross street — Av. Amazonas N34-120 y Av. República —, or the ciudadela with manzana, villa and solar.

HOW IT IS FOUND

COLUMN NAME EC_L1_DIRECCION – Path

80% `direcci[oó]n[a-z0-9_]*` (except when preceded by `mac` , `ip` , `e_?mail` , `web` , `url` , `correo` ; except when followed by `list`
E4) `domicilio[a-z0-9_]*` `dom_?(residencia|particular|laboral|cliente)` `residencia` `lugar_?`
`(de_)?residencia` `calle(_?(principal|secundaria|y_n[uú]mero))?` `address[a-z0-9_]*` (except when
preceded by `mac` , `ip` , `e_?mail` , `web` , `url` ; except when followed by `list` *E5*)

DATA TYPE EC_L1_DIRECCION – Type

Accepts text (length 6 or more); any other type rules the domain out.

VALUE FORMAT EC_L1_DIRECCION - Regex

PATTERN	STRENGTH	NOTES
(?i)(av avenida calle ca pasaje psje callej[oó]n malec[oó]n)\.?\s+[a-záéíóúñ0-9.']{2,40}\s*(n\d{1,2}-\d{1,3} \d{1,5}).*	80%	
(?i).*\b(mz manzana)\.?\s*[a-z0-9]{1,3}\s*(villa solar lote lt)\.?\s*\d{1,3}.*	80%	
(?i).*\b(cdla ciudadela urb urbanizaci[oó]n barrio conjunto lotizaci[oó]n cooperativa)\.?\s+[a-záéíóúñ0-9.']{2,40}.*	70%	
(?i)k(m il[oó]metro)\.?\s*\d{1,3}\s+(v[ií]a carretera autopista).*	70%	

A value that matches none counts 20% against the domain.

HOW IT IS MASKED

The whole line is replaced by a fictitious Ecuadorian address of the same form. Ciudadela, parroquia and postal code have domains of their own for when they live in separate columns.

EC_DIRECCION **SECURE LOOKUP**

Replaces the value with one of the 4,000 values in `ec-direcciones.txt`, chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Av. Amazonas N34-120 y Av. República	→ Malecón Malecón Simón Bolívar 2125 y Gaspar de Villarroel

8.16 Address complement

EC_L1_DIRECCION_COMPLEMENTO

ESSENTIAL COLUMN NAME CHARACTER MAPPING

WHAT IT IS

Manzana, villa, solar, lote, interior, flat, floor, block, stage or building in columns of their own.

HOW IT IS FOUND

COLUMN NAME EC_L1_DIRECCION_COMPLEMENTO - Path

70% manzana mz solar villa(?:n(o|ro|um|umero))? lote lt interior departamento_?(n(o|ro|um|umero)|interior) dpto_?(n(o|ro|um|umero))? piso block bloque etapa edificio n(o|ro|um|umero)_(casa|puerta|dpto|interior) calle_?secundaria intersecci[oó]n referencia_?(de_)?(direcci[oó]n|domicilio) complemento(?:_?direcci[oó]n)?

HOW IT IS MASKED

Character by character, keeping length and format.

EC_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
Mz. B Villa 12	→ Tk. T Tujpu 18

8.17 Bank account

EC_L1_CUENTA_BANCARIA

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

Savings, current and payroll account numbers. The bank code is not personal data and has no domain.

HOW IT IS FOUND

COLUMN NAME EC_L1_CUENTA_BANCARIA – Path

85% cuenta_?(bancaria|banco|corriente|ahorros?
|sueldo|n[oó]mina|abono|dep[ó]sito|destino|origen) n(o|ro|um|umero)?(de_)?(cuenta|cta)
nro_?cta num_?cta cta_?(cte|corriente|ahorros?|bancaria) iban account_?(no|num|number)
bank_?account

HOW IT IS MASKED

Digits are mapped; separators stay.

EC_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
2100123456	→ 1786723060

8.18 Payment card

EC_L1_TARJETA

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Credit and debit card numbers, recognized by the Luhn algorithm.

HOW IT IS FOUND

COLUMN NAME EC_L1_TARJETA – Path

85% tarjeta(_?(cr[eé]dito|d[eé]bito|n(o|ro|um|umero)))? n(o|ro|um|umero)?tarjeta pan card_?
(no|num|number) credit_?card tc_?(numero|num|nro)

VALUE FORMAT EC_L1_TARJETA – Regex

PATTERN	STRENGTH	NOTES
\d{13,19}	90%	LUHN check digit; after removing [\s-]

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Payment Card: another number that passes the Luhn algorithm. Only card-shaped values reach it.

EC_TARJETA REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EC_TARJETA_LUHN PAYMENT CARD

Replaces the card number with another that passes the Luhn check.

EXAMPLES

VALUE

MASKED

4539 5787 6362 1486

→ 1912 8421 0957 2974

8.19 Digital wallet

EC_L1_BILLETERA

COLUMN NAME

VALUE FORMAT

CHARACTER MAPPING

WHAT IT IS

DeUna, Peigo, PayPhone and other wallet identifiers, and the bitcoin addresses some systems keep.

HOW IT IS FOUND

COLUMN NAME

EC_L1_BILLETERA - Path

85%

billetera(_?(digital|electr[oó]nica|m[oó]vil))? wallet(_?(id|address|direccion))? de_?una
peigo payphone dinero_?electr[oó]nico c[oó]digo_?qr qr_?(id|codigo) direcci[oó]n_
(bitcoin|btc|wallet) btc_?(address|direccion)

VALUE FORMAT

EC_L1_BILLETERA - Regex

PATTERN

STRENGTH

NOTES

(bc1|[13])[a-zA-HJ-NP-Z0-9]{25,59}

70%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Character by character, keeping the shape. A masked bitcoin address no longer passes its own checksum, which is what a test copy wants.

EC_CM_ALFANUM

CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

bc1qar0srrr7xfkvy5l643lydnw9re59gtzzwf5mdq

→ mh2wis8tsdm0pdjdv8v511yqxqg4ze53qkjbvp7dgf

8.20 IP address

EC_L1_IP

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

IPv4 and IPv6 addresses from access logs and transactions.

HOW IT IS FOUND

COLUMN NAME

EC_L1_IP - Path

85%

ip ip_?(address|addr|origen|destino|cliente|usuario|acceso|login|remota|publica)
direcci[oó]n_?ip dir_?ip ipv[46] remote_?addr(ess)? client_?ip host_?ip

VALUE FORMAT EC_L1_IP - Regex

PATTERN	STRENGTH	NOTES
((25[0-5] 2[0-4]\d 1?\d?\d)\.){3}(25[0-5] 2[0-4]\d 1?\d?\d)	100%	
[0-9A-Fa-f]{1,4}(:[0-9A-Fa-f]{0,4}){2,7}	90%	

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Each IPv4 octet is mapped to 1-254; an IPv6 address is mapped keeping its hex characters.

EC_IP REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: EC_CM_HEX .

EC_OCTETO NUMERIC MAPPING

Maps the number to another between 1 and 254.

EC_CM_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
186.42.163.10	→ 153.35.221.16

8.21 Device identifier

EC_L1_DISPOSITIVO

COLUMN NAME **VALUE FORMAT** **CHARACTER MAPPING**

WHAT IT IS

IMEI, IMSI, ICCID, MAC address, advertising and session identifiers.

HOW IT IS FOUND

COLUMN NAME EC_L1_DISPOSITIVO - Path

80% imei imsi iccid mac mac_?address direcci[oó]n_?mac device_(id|uuid|serial) id_?
(dispositivo|equipo|celular) serial_(equipo|celular|dispositivo) advertising_?id idfa
gaid session_?id id_?sesi[oó]n

VALUE FORMAT EC_L1_DISPOSITIVO - Regex

PATTERN	STRENGTH	NOTES
([0-9A-Fa-f]{2}[:-]){5}[0-9A-Fa-f]{2}	100%	
\d{15}	80%	LUHN check digit

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Character by character, with hex digits kept hex: a MAC stays a MAC.

EC_CM_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
3c:22:fb:9a:10:e4	→ 5e:83:bb:3c:31:f9

8.22 Cookies and tracking identifiers

EC_L1_COOKIE

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

The identifier a site stores in the browser and the campaign parameters that travel with it. Article 4 defines the **elaboración de perfiles** these feed: any processing that evaluates or predicts a person's professional performance, economic situation, health, preferences, interests, location or movement.

HOW IT IS FOUND

COLUMN NAME EC_L1_COOKIE - Path

80% `cookie(_?(id|value|valor|nombre))?` `cookies` `_?ga` `_?gid` `fbp` `fbclid` `utm_?`
`(source|medium|campaign|term|content)` `id_?(navegador|browser|visitante)` `visitor_?id`
`tracking_?id` `client_?id`

HOW IT IS MASKED

Character by character, keeping length and format.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
GA1.2.123456789.1700000000	→ ZE0.5.891552838.7889916184

8.23 Vehicle plate

EC_L1_PLACA

COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

The plate with three letters and three or four digits, PCA-1234 ; the first letter is the province — A Azuay, G Guayas, P Pichincha, M Manabí.

HOW IT IS FOUND

COLUMN NAME EC_L1_PLACA - Path

85% `placa(_?(veh[ií]culo|carro|moto|automotor))?` `n(o|ro|um|umero)_?placa` `matr[ií]cula_?`
`(veh[ií]culo|automotor)` `license_?plate` `plate(_?(no|number))?`

VALUE FORMAT EC_L1_PLACA - Regex

PATTERN	STRENGTH	NOTES
<code>[A-Z]{3}[\s\~]?[d]{3,4}</code>		70%
<code>[A-Z]{2}[\s\~]?[d]{3,4}</code>		40%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Letters to letters and digits to digits, position by position: the plate format stays.

EC_CM_DIGITOS_LETRAS CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

PCA-1234

→

MQP-8512

GSB-123

→

WKG-104

8.24 Vehicle and licence

EC_L1_VEHICULO

COLUMN NAME

VALUE FORMAT

CHARACTER MAPPING

WHAT IT IS

The VIN or chassis number, the engine number, the registration and the driving licence.

HOW IT IS FOUND

COLUMN NAME

EC_L1_VEHICULO - Path

85%

```
vin    chasis    n(o|ro|um|umero)?(chasis|motor|serie|vin)    motor_?n(o|ro|um|umero)
matr[ií]cula(_?veh[ií]culo)?    licencia_?(de_)?conducir(_?(n(o|ro|um|umero)|tipo))?
n(o|ro|um|umero)_?licencia    puntos_?licencia
```

VALUE FORMAT

EC_L1_VEHICULO - Regex

PATTERN

[A-HJ-NPR-Z0-9]{17}

STRENGTH

NOTES

70%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Character by character, keeping length and format.

EC_CM_ALFANUM

CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

8AJHA8CD3J1234567

→

8EWKE1QN1T0772894

8.25 Property

EC_L1_INMUEBLE

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

The cadastral key, the predial code and the entry in the municipal Registro de la Propiedad: they lead to the home and its owner.

HOW IT IS FOUND

COLUMN NAME

EC_L1_INMUEBLE - Path

80%

```
clave_?catastral    c[ó]digo_?(catastral|predial|de_?predio)    n(o|ro|um|umero)_?(de_)?
(predio|inmueble|catastral)    catastro    ficha_?catastral    inscripci[ó]n_?(registral|de_?la_?
propiedad)    registro_?(de_?la_?)?propiedad    repertorio    n(o|ro|um|umero)_?(de_?)?medidor
```

HOW IT IS MASKED

Character by character, keeping length and format.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

12-345-678

MASKED

→ 46-916-122

8.26 Customer and contract numbers

EC_L1_CONTRATO

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

Contract, policy, loan, application, member, subscriber, supply, meter, case or administrative file numbers, and customer or employee codes.

HOW IT IS FOUND

COLUMN NAME

EC_L1_CONTRATO - Path

70%

```
n(o|ro|um|umero)?(contrato|p[ó]liza|cr[eé]dito|pr[eé]stamo|solicitud|expediente_?
(administrativo|interno)|tr[aá]mite|referencia|cliente|socio|suscriptor|suministro|medidor|ser
vicio|caso|ticket|beneficio) contrato?(n(o|ro|um|umero)) c[ó]digo_?
(cliente|socio|empleado|afiliado|suministro) id?(cliente|socio|empleado|afiliado)
n(o|ro)?empleado c[ó]digo?(de_)?(empleado|trabajador|n[ó]mina)
```

HOW IT IS MASKED

Character by character, keeping length and format. The mapping is deterministic, so joins keep working.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

CT-123456

MASKED

→ NG-557411

8.27 User and social media

EC_L1_USUARIO

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

Login, alias and social media profiles.

HOW IT IS FOUND

COLUMN NAME

EC_L1_USUARIO - Path

70%

```
usuario user_?name username login alias apodo nick(_?name)? perfil_?
(instagram|facebook|twitter|tiktok|linkedin|x) instagram facebook twitter tiktok
linkedin red_?social handle
```

HOW IT IS MASKED

Character by character, keeping length and format.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

@narcisa_zambrano

MASKED

→ @dipcola_gojmyeju

8.28 Credentials

EC_L1_CREDENTIAL

COLUMN NAME

REGEX DECOMPOSE

WHAT IT IS

Passwords and their hashes, PIN, security and OTP codes, CVV, tokens, API keys, the key of an electronic signature, private keys and seed phrases.

HOW IT IS FOUND

COLUMN NAME

EC_L1_CREDENTIAL - Path

90%

contrase[nñ]a clave *(except when followed by catastral, primaria, for[aá]nea, valor, producto)*
password passwd pwd pass hash_?(clave|contrasena|password) pin c[oó]digo_
(seguridad|verificaci[oó]n|acceso|otp) cvv cvc token(_?(acceso|refresh|api))? access_
token refresh_?token api_?key llave_?(api|privada) secreto secret otp firma_
electr[oó]nica_?(clave|pin) pregunta_?secreta respuesta_?secreta frase_?semilla seed_
phrase

HOW IT IS MASKED

Every character becomes X. A credential has no use in a test copy, and a hash masked character by character could still be cracked.

EC_REDACTAR

REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE

Clave#2024

MASKED

→ XXXXXXXXXX

8.29 Geolocation

EC_L1_GEOLOCALIZACION

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

Latitude and longitude, together or in separate columns, from apps, visits, addresses and deliveries.

HOW IT IS FOUND

COLUMN NAME

EC_L1_GEOLOCALIZACION - Path

85%

lat latitud lon lng longitud coordenadas? coord_?[xy] geo_
(lat|lon|lng|loc|localizacion|posicion) georreferenciaci[oó]n geolocalizaci[oó]n gps(_?
(ubicaci[oó]n|posici[oó]n|coordenadas))? ubicaci[oó]n_?gps

50%

long

VALUE FORMAT EC_L1_GEOLOCALIZACION – Regex

PATTERN	STRENGTH	NOTES
-?[0-5]\.\d{3,}\s*,\s*-(7[5-9] 8[01] 9[01])\.\d{3,}	90%	
-(7[5-9] 8[01])\.\d{4,}	50%	
-?[0-5]\.\d{4,}	40%	

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The integer part and first decimal stay — about 11 km — and the other decimals are mapped: the place is generalized, not moved.

EC_COORDENADA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_CM_DIGITOS **CHARACTER MAPPING**

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
-0.180653,-78.467834	→ -0.134127,-78.499081

8.30 Free text

EC_L1_TEXTO_LIBRE

COLUMN NAME **DATA TYPE** **VALUE FORMAT** **FREE TEXT REDACTION**

WHAT IT IS

Remarks, complaint descriptions, accounts of events and messages where anything gets written — cédula, RUC, phone and names in the middle of a sentence.

HOW IT IS FOUND

COLUMN NAME EC_L1_TEXTO_LIBRE – Path

60% observaci[oó]n(es)? obs comentarios? notas? anotaci[oó]n(es)? glosa sustento
descripci[oó]n?(queja|reclamo|solicitud|caso|hechos|novedad|atenci[oó]n|denuncia) detalle_
(reclamo|solicitud|caso|atenci[oó]n) hechos relato justificaci[oó]n mensaje *(except when
preceded by list E6; except when followed by list E7)* texto(?:libre)? resumen?(caso|atenci[oó]n) queja
reclamo denuncia_?texto remarks? comments? notes? free_?text memo

DATA TYPE EC_L1_TEXTO_LIBRE – Type

Accepts text (length 30 or more); any other type rules the domain out.

VALUE FORMAT EC_L1_TEXTO_LIBRE – Regex

PATTERN	STRENGTH	NOTES
(0[1-9] 1\d 2[0-4] 30)[0-5]\d{7}	70%	anywhere in the value
[\w.+\\-]+@[\\w\\-]+\\. [A-Za-z]{2,}	70%	anywhere in the value
0?9\d{2}[\\s\\-]?\\d{3}[\\s\\-]?\\d{3}	60%	anywhere in the value

A value that matches none does not count against the domain.

HOW IT IS MASKED

Free Text Redaction: cédulas, RUCs, e-mail, mobiles, cards, plates and IPs become [DATO] , and the names of the lists [NOMBRE] .

EC_TEXTO_LIBRE FREE TEXT REDACTION

Replaces the text matching 8 patterns with [DAT0] , and the 1,668 terms of ec-texto-nombres.txt with [NOMBRE] .

EXAMPLES

VALUE

Cliente Zambrano, cédula 1710034065, correo n.zambrano@gmail.com, cel 0991234567.

→

MASKED

Cliente [NOMBRE], cédula [DAT0], correo [DAT0], cel [DAT0].

Quasi-identifiers

Data that identifies nobody on its own, but does in combination. Ecuador is divided into about 1,040 parroquias, and 913 of them had fewer than 20,000 inhabitants in the 2022 census. A birth date, a sex and one of those parroquias point at a handful of people. These attributes are generalized or shifted rather than replaced, and the data keeps its analytical value.

9.1 Birth date

EC_L2_FECHA_NACIMIENTO

ESSENTIAL COLUMN NAME DATA TYPE DATE SHIFT

WHAT IT IS

The birth date, as a date, timestamp or text. It also decides whether the row belongs to a child or adolescent, a special category under art. 25 b).

HOW IT IS FOUND

COLUMN NAME EC_L2_FECHA_NACIMIENTO - Path

90% (fecha|fec|fch|f)?(de_)?nac(imiento)? fnac fechanac(imiento)? cumplea[nños]os dob
date_of_birth birth_date birthday

DATA TYPE EC_L2_FECHA_NACIMIENTO - Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the domain out.

HOW IT IS MASKED

Shifted up to 60 days either way. Ages and school years change for few people; the date stays plausible and the same birth date always moves the same way.

EC_FECHA_NACIMIENTO DATE SHIFT

Moves the date by -60 to 60 days.

EXAMPLES

VALUE		MASKED
1985-07-20	→	1985-08-05T00:00

9.2 Birth year

EC_L2_ANIO_NACIMIENTO

COLUMN NAME DATA TYPE REGEX DECOMPOSE

WHAT IT IS

The birth year in a column of its own.

HOW IT IS FOUND

COLUMN NAME EC_L2_ANIO_NACIMIENTO - Path

90% a[nño]o?(de_)?nac(imiento)? anio_nac(imiento)? year_of_birth birth_year yob

DATA TYPE EC_L2_ANIO_NACIMIENTO - Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the domain out.

HOW IT IS MASKED

The decade stays and the last digit is mapped: 1985 becomes a year from 1980 to 1989.

EC_ANIO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EC_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
1985	→	1980

9.3

Age

EC_L2_EDAD

COLUMN NAME DATA TYPE REGEX DECOMPOSE

WHAT IT IS

Age in years, also age group or five-year band.

HOW IT IS FOUND

COLUMN NAME EC_L2_EDAD – Path

85% edad(_?(actual|a[nñ]os|anios|paciente|afiliado|cliente|al_?
(ingreso|diagn[ó]stico|fallecimiento)))? grupo_?etario rango_?(de_?)?edad quinquenio
age(_?(years|group)))?

DATA TYPE EC_L2_EDAD – Type

Accepts number (length up to 6) or text (length up to 6); any other type rules the domain out.

HOW IT IS MASKED

The tens digit stays and the unit is mapped (37 becomes 30–39). A hundred or more becomes 90: rare ages identify.

EC_EDAD REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
37	→	35
104	→	90

9.4

Personal event dates

EC_L2_FECHA_EVENTO

COLUMN NAME DATA TYPE DATE SHIFT

WHAT IT IS

Dates that say something about the person: death, marriage, hiring and leaving, retirement, admission, diagnosis, arrest, sentence, sick leave, cédula issue.

HOW IT IS FOUND

COLUMN NAME EC_L2_FECHA_EVENTO – Path

80% (fecha|fec|fch|f)?(de_)?
(defunci[oó]n|fallecimiento|muerte|matrimonio|divorcio|ingreso|salida|egreso|alta|baja|retiro|
contrataci[oó]n|despido|jubilaci[oó]n|hospitalizaci[oó]n|internamiento|diagn[oó]stico|atenci[o
ó]n|consulta|parto|detenci[oó]n|sentencia|expedici[oó]n(_?(cedula|documento))?)|caducidad_
c[éé]dula|vacunaci[oó]n|cirug[ií]a|accidente|reposo) fec_
(exp|expedicion|defuncion|ingreso|salida|baja) date_?of_
(death|marriage|hire|admission|discharge)

DATA TYPE EC_L2_FECHA_EVENTO – Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the domain out.

HOW IT IS MASKED

Shifted up to 30 days either way.

EC_FECHA_EVENTO DATE SHIFT

Moves the date by -30 to 30 days.

EXAMPLES

VALUE

MASKED

2021-03-15

→

2021-03-05T00:00

9.5

Sex

EC_L2_SEX0

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Sex as the system records it: F / M , Femenino / Masculino , Mujer / Hombre or the codes 1 and 2.

HOW IT IS FOUND

COLUMN NAME EC_L2_SEX0 – Path

85% sexo(_?(biol[oó]gico|al_?nacer|paciente|afiliado|cliente))? (tp|tipo|cod|cd|id)?sexo
g[eé]nero (except when followed by *identidad* , *autopercebido*) sex gender (except when followed by
identity)

KNOWN VALUES EC_L2_SEX0 – List

LIST

VALUES

STRENGTH

FIRST VALUES

ec-detectar-sexo.txt

9

50%

f, m, h, femenino, masculino, mujer

A value that matches none counts 50% against the domain.

HOW IT IS MASKED

Replaced within its own vocabulary: F stays F or M , 1 stays 1 or 2 . The **identidad de género** of art. 4 has its own domain among the sensitive data.

EC_SEX0 REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_SEX0_FEMENINO_MASCULINO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-sexo-femenino-masculino.txt , chosen by a keyed hash.

EC_SEX0_MUJER_HOMBRE SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-sexo-mujer-hombre.txt` , chosen by a keyed hash.

EC_SEX0_FEMALE_MALE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-sexo-female-male.txt` , chosen by a keyed hash.

EC_SEX0_F_M **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-sexo-f-m.txt` , chosen by a keyed hash.

EC_SEX0_H_M **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-sexo-h-m.txt` , chosen by a keyed hash.

EC_SEX0_1_2 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-sexo-1-2.txt` , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
F	→	M
Femenino	→	Femenino

9.6 Parroquia and cantón

EC_L2_PARROQUIA

COLUMN NAME

KNOWN VALUES

DATA CLEANSING

WHAT IT IS

The parroquia or cantón of residence, birth or care, by name — alone or with the cantón or the provincia, `Facundo Vela – Bolívar` .

HOW IT IS FOUND

COLUMN NAME

EC_L2_PARROQUIA – Path

85% parroquia(_?(residencia|nacimiento|domicilio))? (except when followed by `cod` , `codigo` , `id` , `eclesiastica`) cant[ó]n(_?(residencia|nacimiento|domicilio))? (except when followed by `cod` , `codigo` , `id`) canton (except when followed by `cod` , `codigo` , `id`) ciudad(_?(residencia|nacimiento|domicilio|cliente))? (nom|nombre|desc)?_(parroquia|canton|ciudad) lugar?(de_)?nacimiento city town

KNOWN VALUES

EC_L2_PARROQUIA – List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-parroquias.txt	1,160	80%	Guaranda, Facundo Vela, Julio E. Moreno (Catanahuan Grande), San Lorenzo, San Simón (Yacoto), San Simon (Yacoto)

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A parroquia with fewer than 20,000 inhabitants becomes the nearest parroquia of its cantón with at least 20,000 — or of its provincia when the cantón has none. The 93 cantones under 20,000 follow the same rule inside their provincia. The 24 provincias stay.

EC_PARROQUIA **DATA CLEANSING**

Replaces each value found in `ec-parroquias-generalizadas.txt` (25,214 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

VALUE		MASKED
Chavezpamba	→	Pomasqui
Quito	→	Quito
Facundo Vela – Bolívar	→	Guaranda – Bolívar
Isabela	→	Santa Cruz

9.7 DPA code

EC_L2_DPA

COLUMN NAME

VALUE FORMAT

DATA CLEANSING

WHAT IT IS

The código de la División Política Administrativa: two digits for the provincia, four for the cantón and six for the parroquia — the same six digits the postal code uses.

HOW IT IS FOUND

COLUMN NAME

EC_L2_DPA – Path

85%

```
dpa[a-z0-9_]* (cod|codigo|cd|id)?(dpa|parroquia|canton|cant|ciudad|lugar)(_?
(inei|inec|censo|res|residencia|nac|nacimiento))? (cod|codigo)?
(inec|geogr[aá]fico|divisi[oó]n?pol[íi]tica) codparroquia codcanton
```

VALUE FORMAT

EC_L2_DPA – Regex

PATTERN

```
(0[1-9]|1\d|2[0-4]|90)\d{2}(\d{2})?
```

STRENGTH

NOTES

40%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The same generalization as the name: a small parroquia's code becomes the code of the parroquia that replaces its name. The code of a large one does not change.

EC_DPA DATA CLEANSING

Replaces each value found in `ec-dpa-generalizado.txt` (1,006 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

VALUE		MASKED
170158	→	170177
170150	→	170150

9.8 Ciudadela, urbanización and barrio

EC_L2_CIUDELA

COLUMN NAME

SECURE LOOKUP

WHAT IT IS

The ciudadela, urbanización, cooperativa, conjunto, recinto, comuna or barrio of residence — the unit below the parroquia, and the one a delivery address really uses.

HOW IT IS FOUND

COLUMN NAME EC_L2_CIUDELA - Path

80% ciudadela cdla urbanizaci[oó]n urb barrio(_?(residencia|domicilio))? lotizaci[oó]n cooperativa_(de_)?vivienda conjunto_(habitacional|residencial) recinto comuna sector(_?(vivienda|domicilio|residencia))? (nom|nombre)?(ciudadela|barrio|urbanizacion) neighbou?rhood

HOW IT IS MASKED

Replaced by a common Ecuadorian one, unrelated to the original.

EC_CIUDELA **SECURE LOOKUP**

Replaces the value with one of the 108 values in ec-ciudadelas.txt, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Cdla. Kennedy

→

Cdla. Los Almendros

9.9 Postal code

EC_L2_CODIGO_POSTAL

COLUMN NAME **DATA TYPE** **REGEX DECOMPOSE**

WHAT IT IS

The six-digit postal code in use since 2007: it is the DPA code, so the first two digits are the provincia.

HOW IT IS FOUND

COLUMN NAME EC_L2_CODIGO_POSTAL - Path

85% c[oó]digo_?postal cod_?postal codpostal zip(_?code)? postal_?code cp

DATA TYPE EC_L2_CODIGO_POSTAL - Type

Accepts text (length 5 to 6) or number (length up to 6); any other type rules the domain out.

HOW IT IS MASKED

The first two digits stay and the others become zeros: 170150 becomes 170000.

EC_CODIGO_POSTAL **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE

MASKED

170150

→

170000

9.10 Marital status

EC_L2_ESTADO_CIVIL

COLUMN NAME **KNOWN VALUES** **REGEX DECOMPOSE**

WHAT IT IS

Single, married, in a union of fact — which the Constitution puts on the same footing as marriage —, separated, divorced or widowed, in words or as a code.

HOW IT IS FOUND

COLUMN NAME EC_L2_ESTADO_CIVIL – Path

85% estado_?civil est_?civil (cod|tipo|id)_?estado_?civil situaci[oó]n_?conyugal uni[oó]n_?de_?hecho marital_?status civil_?status

KNOWN VALUES EC_L2_ESTADO_CIVIL – List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-estado-civil.txt	22	70%	Soltero/a, Casado/a, Unión de hecho, Union de hecho, Separado/a, Divorciado/a

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by another status from the list; a code stays a code.

EC_ESTADO_CIVIL **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_ESTADO_CIVIL_LISTA .

EC_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-si-no.txt , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-true-false.txt , chosen by a keyed hash.

EC_BANDERA_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-yes-no.txt , chosen by a keyed hash.

EC_BANDERA_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-s-n.txt , chosen by a keyed hash.

EC_BANDERA_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-y-n.txt , chosen by a keyed hash.

EC_BANDERA_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-0-1.txt , chosen by a keyed hash.

EC_ESTADO_CIVIL_CODIGO **SECURE LOOKUP**

Replaces the value with one of the 6 values in ec-estado-civil-codigos.txt , chosen by a keyed hash.

EC_ESTADO_CIVIL_LISTA **SECURE LOOKUP**

Replaces the value with one of the 6 values in ec-estado-civil.txt , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Unión libre	→ Divorciado/a

9.11 Occupation

EC_L2_OCUPACION

COLUMN NAME **REGEX DECOMPOSE**

WHAT IT IS

Occupation, profession or trade, in words or by the CIUO code. A rare occupation in a small parroquia identifies on its own.

HOW IT IS FOUND

COLUMN NAME	EC_L2_OCUPACION – Path
80%	ocupaci[oó]n profes[ió]n oficio ciuo (cod codigo)?(ocupacion profesion ciuo) puesto?(de_)?trabajo categor[ií]a?ocupacional cargo?(actual empleado) occupation profession job?title
50%	cargo puesto

HOW IT IS MASKED

Text is replaced by a common occupation; a code is mapped digit by digit.

EC_OCUPACION_0_CODIGO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: EC_OCUPACION .

EC_CM_DIGITOS	CHARACTER MAPPING
Maps each character to another of its group (1 groups), keeping length and format.	
EC_OCUPACION	SECURE LOOKUP
Replaces the value with one of the 38 values in ec-ocupaciones.txt , chosen by a keyed hash.	

EXAMPLES

VALUE		MASKED
Gerente de operaciones	→	Contador
2221	→	1980

9.12 Employer

EC_L2_EMPLEADOR

COLUMN NAME	SECURE LOOKUP
-------------	---------------

WHAT IT IS

The name of the company or institution where the person works.

HOW IT IS FOUND

COLUMN NAME	EC_L2_EMPLEADOR – Path
90%	empleador(_?(nombre raz[oó]n?social ruc))? (nombre nom)?(empleador empresa?donde? trabaja patrono) empresa?(donde_)?(trabaja labora) lugar?(de_)?trabajo centro?(de_)? (trabajo labores) entidad?empleadora employer(_?name)? workplace

HOW IT IS MASKED

Replaced by a fictitious company.

EC_EMPLEADOR SECURE LOOKUP

Replaces the value with one of the 24 values in ec-empleadores.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Corporación Comercial Ecuatoriana S.A.	→	Cooperativa Agrícola La Unión

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

The level of instruction attained — from none to a postgraduate degree — in the categories the census uses, in words or as a code.

HOW IT IS FOUND

COLUMN NAME

EC_L2_NIVEL_INSTRUCCION - Path

80%

escolaridad nivel_?(educativo|de_?estudios|acad[eé]mico|alcanzado|de_?instrucci[oó]n) instrucci[oó]n(_?formal)? grado_?(de_?)?(estudio|instrucci[oó]n|escolaridad) m[aá]ximo_? nivel a[nñ]os_?(de_?)?estudio education(_?level)?

KNOWN VALUES

EC_L2_NIVEL_INSTRUCCION - List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-nivel-instruccion.txt	25	60%	Ninguno, Centro de alfabetización, Centro de alfabetizacion, Educación inicial, Educacion inicial, Educación general básica

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by another level from the list; a code stays a code.

EC_NIVEL_INSTRUCCION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_NIVEL_INSTRUCCION_LISTA .

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-si-no.txt , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-true-false.txt , chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-yes-no.txt , chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-s-n.txt , chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-y-n.txt , chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-0-1.txt , chosen by a keyed hash.

EC_NIVEL_INSTRUCCION_CODIGO SECURE LOOKUP

Replaces the value with one of the 8 values in ec-nivel-instruccion-codigos.txt , chosen by a keyed hash.

EC_NIVEL_INSTRUCCION_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in ec-nivel-instruccion.txt , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Educación superior en curso	→ Bachillerato

9.14 School

EC_L2_INSTITUCION_EDUCATIVA

COLUMN NAME SECURE LOOKUP

WHAT IT IS

The unidad educativa, institute or university, by name or by the Ministry of Education AMIE code.

HOW IT IS FOUND

COLUMN NAME EC_L2_INSTITUCION_EDUCATIVA – Path

75% unidad_?educativa instituci[oó]n_?educativa centro_?(educativo|de_?estudios)
(nombre|nom)_?(institucion_?educativa|escuela|colegio|universidad) escuela colegio
instituto universidad c[oó]digo_?amie amie school(_?name)? university

HOW IT IS MASKED

Replaced by a fictitious school.

EC_INSTITUCION_EDUCATIVA SECURE LOOKUP

Replaces the value with one of the 12 values in `ec-instituciones-educativas.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Universidad Central del Ecuador

→

Escuela Politécnica del Austro

9.15 Dependants and children

EC_L2_PERSONAS_A_CARGO

COLUMN NAME DATA TYPE REGEX DECOMPOSE

WHAT IT IS

Number of dependants, children, derechohabientes or household members.

HOW IT IS FOUND

COLUMN NAME EC_L2_PERSONAS_A_CARGO – Path

80% personas_?a_?cargo cargas_?familiares n(o|ro|um|umero)_?(de_?)?
(hijos|dependientes|cargas|derechohabientes|personas_?a_?cargo) cant(idad)?_
(hijos|dependientes|personas_?hogar|miembros) hijos dependientes derechohabientes
personas_?(en_?el_?)?hogar tama[ñ]o_?(de_l_?)?hogar miembros_?(de_l_?)?hogar number_?of_?
children dependents

DATA TYPE EC_L2_PERSONAS_A_CARGO – Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the domain out.

HOW IT IS MASKED

Values from 0 to 5 stay; above 5 they become 5. Very large households are rare, and a rare count identifies.

EC_CONTEO_LIMITADO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE

MASKED

9

→

5

Sensitive data (art. 4)

The list of article 4 is the longest in the region, and three of its items appear in no neighbouring law: the **identidad cultural**, the **condición migratoria** and the **pasado judicial**. Health has a group of its own below, and the judicial record the PENAL group; everything else is here, together with the trade union membership and the victim status the open clause of the same article reaches.

10.1 Ethnicity

EC_L3_ETNIA

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Self-identification as mestizo, montubio, afroecuatoriano, negro, mulato, indígena or blanco — the categories the 2022 census asked. Article 4 names the etnia first among the sensitive data.

HOW IT IS FOUND

COLUMN NAME	EC_L3_ETNIA – Path		
95%	etnia origen_?([eé]tnico racial) grupo_?[eé]tnico pertenencia_?[eé]tnica autoidentificaci[oó]n(_?[eé]tnica)? autodefinici[oó]n (cod codigo tipo id)? (etnia autoidentificacion) raza ind[ií]lgena montubio afroecuatoriano afrodescendiente ethnicity race		
KNOWN VALUES	EC_L3_ETNIA – List		
LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-etnia.txt	20	90%	Mestizo, Montubio, Afroecuatoriano, Negro, Mulato, Indígena
ec-detectar-codigos-banderas.txt	11	30%	s, n, sí, si, no, 1

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A flag stays a flag, a code is replaced by another code and any other value by a category from the list.

EC_AUTOIDENTIFICACION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_AUTOIDENTIFICACION_LISTA .

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-si-no.txt , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-true-false.txt , chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-yes-no.txt , chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-s-n.txt , chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-y-n.txt , chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.

EC_AUTOIDENTIFICACION_CODIGO **SECURE LOOKUP**

Replaces the value with one of the 8 values in `ec-autoidentificacion-codigos.txt` , chosen by a keyed hash.

EC_AUTOIDENTIFICACION_LISTA **SECURE LOOKUP**

Replaces the value with one of the 8 values in `ec-autoidentificacion.txt` , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Indígena	→	Otro
S	→	S

10.2 Cultural identity

EC_L3_IDENTIDAD_CULTURAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The **identidad cultural** article 4 names apart from the etnia: the indigenous nationality or pueblo a person belongs to. The Constitution recognizes fourteen nationalities and eighteen kichwa pueblos, and naming one of them names a few thousand people. No other law of the region lists this.

HOW IT IS FOUND

COLUMN NAME EC_L3_IDENTIDAD_CULTURAL - Path

90% `identidad_?cultural nacionalidad_?(ind[ií]gena|originaria) pueblo_?`
`(ind[ií]gena|originario|al_?que_?pertenece) pertenece_?(a_?)?`
`(pueblo|nacionalidad|comunidad) comunidad_?(ind[ií]gena|ancestral) nacionalidad_?`
`(ecuatoriana)? (except when followed by pais , extranjera , documento ) indigenous_?people`

KNOWN VALUES EC_L3_IDENTIDAD_CULTURAL - List

LIST	VALUES	STRENGTH	FIRST VALUES
<code>ec-detectar-nacionalidades-indigenas.txt</code>	67	85%	Kichwa, Shuar, Achuar, Awá, Awa, Chachi

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by another nationality or pueblo from the list; a flag stays a flag.

EC_IDENTIDAD_CULTURAL **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_IDENTIDAD_CULTURAL_LISTA .

EC_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt` , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt` , chosen by a keyed hash.

EC_BANDERA_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt` , chosen by a keyed hash.

EC_BANDERA_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.

EC_BANDERA_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt` , chosen by a keyed hash.

EC_BANDERA_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

EC_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EC_IDENTIDAD_CULTURAL_LISTA **SECURE LOOKUP**

Replaces the value with one of the 30 values in `ec-identidad-cultural.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Pueblo Manta-Huancavilca

→ **Siekopai**

10.3 Language

EC_L3_LENGUA

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The mother or ancestral language. Thirteen indigenous languages are spoken in Ecuador beside Spanish, and some by a few hundred people: the language says who a person is.

HOW IT IS FOUND

COLUMN NAME

EC_L3_LENGUA - Path

90%

```
lengua(_?(ind[ii]gena|materna|nativa|originaria|ancestral|de_?se[nñas]))? idioma(_?(ind[ii]gena|materno|nativo|ancestral))? habla_?(lengua|idioma) mother_?tongue native_?language
```

KNOWN VALUES

EC_L3_LENGUA - List

LIST

VALUES STRENGTH FIRST VALUES

ec-detectar-lenguas.txt	29	80%	Castellano, Kichwa, Shuar chicham, Achuar chicham, Awapit, Cha'palaa
-------------------------	----	-----	--

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by a language from the list; a flag stays a flag.

EC_LENGUA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_LENGUA_LISTA`.

EC_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

EC_BANDERA_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

EC_BANDERA_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

EC_BANDERA_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

EC_BANDERA_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EC LENGUA_LISTA SECURE LOOKUP

Replaces the value with one of the 16 values in `ec-lenguas.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Kichwa de la sierra

→

Otra

10.4 Migratory status

EC_L3_CONDICION_MIGRATORIA

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The **condición migratoria** article 4 lists among the sensitive data, and the stateless and refugee status the same article names: residence category, refugee or asylum application, regular or irregular status, human mobility. In Ecuador this column separates the nationals from the Venezuelan and Colombian communities.

HOW IT IS FOUND

COLUMN NAME

EC_L3_CONDICION_MIGRATORIA - Path

95%

condici[oó]n?migratoria estatus?migratorio situaci[oó]n?migratoria categor[ií]a?
migratoria movilidad?humana refugiad[oa] solicitante?(de_?)(refugio|asilo)
asilad[oa] ap[aá]trida protecci[oó]n?internacional migrante desplazad[oa]
retornad[oa] deportad[oa] migratory?status refugee

KNOWN VALUES

EC_L3_CONDICION_MIGRATORIA - List

LIST

VALUES

STRENGTH

FIRST VALUES

ec-detectar-condicion-
migratoria.txt

22

80%

Ecuatoriano, Residente permanente, Residente temporal, Visitante
temporal, Refugiado, Solicitante de refugio

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by another status from the list; a flag stays a flag.

EC_CONDICION_MIGRATORIA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_CONDICION_MIGRATORIA_LISTA .

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EC_CONDICION_MIGRATORIA_LISTA SECURE LOOKUP

Replaces the value with one of the 10 values in `ec-condicion-migratoria.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Residente en trámite

→

En situación irregular

10.5 Nationality

EC_L3_NACIONALIDAD

COLUMN NAME

KNOWN VALUES

SECURE LOOKUP

WHAT IT IS

The nationality of a country and the country of birth or origin. Article 4 does not name it, but it travels with the migratory status it does name.

HOW IT IS FOUND

COLUMN NAME

EC_L3_NACIONALIDAD - Path

85%

nacionalidad_?(pais|extranjera|documento) pa[ií]s_?(de_)?
(nacimiento|origen|nacionalidad|procedencia) ciudadan[ií]a nationality citizenship
country_of_birth

KNOWN VALUES

EC_L3_NACIONALIDAD - List

LIST

VALUES

STRENGTH

FIRST VALUES

VALUES	STRENGTH	FIRST VALUES
ec-detectar-nacionalidades.txt	32	70%

Ecuatoriana, Venezolana, Colombiana, Peruana, Cubana, Haitiana

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by a nationality from the list: the Ecuadorian one and the largest foreign communities.

EC_NACIONALIDAD_PAIS SECURE LOOKUP

Replaces the value with one of the 14 values in `ec-nacionalidades.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Panameña

→

Boliviana

10.6 Religion

EC_L3_RELIGION

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The religion, church or congregation the person belongs to.

HOW IT IS FOUND

COLUMN NAME EC_L3_RELIGION – Path

95%
religi[oó]n
creencia_?(religiosa|espiritual)
convicci[oó]n_?(religiosa|espiritual)
credo
confesi[oó]n_?religiosa
culto
iglesia(_?(a_?la_?que_?pertenece|nombre))?
denominaci[oó]n_?religiosa
(cod|codigo|tipo)_?religion
religion
church

KNOWN VALUES EC_L3_RELIGION – List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-religiones.txt	35	90%	Católica, Catolica, Evangélica, Evangelica, Testigo de Jehová, Testigo de Jehova

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced by a religion from the list.

EC_RELIGION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_RELIGION_LISTA .

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-si-no.txt , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-true-false.txt , chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-yes-no.txt , chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-s-n.txt , chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-y-n.txt , chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-0-1.txt , chosen by a keyed hash.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EC_RELIGION_LISTA SECURE LOOKUP

Replaces the value with one of the 10 values in ec-religiones.txt , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Ortodoxa	→ Otra

10.7 Ideology

EC_L3_IDEOLOGIA

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The **ideología** article 4 names on its own, beside the filiación política: leaning, sympathy, voting intention, and the philosophical or moral convictions of the same family.

HOW IT IS FOUND

COLUMN NAME EC_L3_IDEOLOGIA - Path

90% ideolog[ií]a(_?pol[ií]tica)? opini[oó]n_?pol[ií]tica orientaci[oó]n_?pol[ií]tica
tendencia_?pol[ií]tica preferencia_?(pol[ií]tica|partidaria|electoral) intenci[oó]n_?
(de_)?voto simpat[ií]a_?pol[ií]tica convicci[oó]n_?(filos[oó]fica|moral) political_?
(opinion|orientation|view) voting_?intention

KNOWN VALUES EC_L3_IDEOLOGIA - List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-ideologias.txt	16	70%	Izquierda, Centroizquierda, Centro, Centroderecha, Derecha, Independiente

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by a position from the list — left to right, independent.

EC_IDEOLOGIA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_IDEOLOGIA_LISTA .

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-si-no.txt , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-true-false.txt , chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-yes-no.txt , chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-s-n.txt , chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-y-n.txt , chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-0-1.txt , chosen by a keyed hash.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EC_IDEOLOGIA_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in ec-ideologias.txt , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Oficialista	→ Centro

10.8 Political affiliation

EC_L3_FILIACION_POLITICA

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The political organization the person belongs to, from the register of the Consejo Nacional Electoral.

HOW IT IS FOUND

COLUMN NAME EC_L3_FILIACION_POLITICA - Path

95% partido(_?pol[ií]tico)?(_?(afiliaci[oó]n|nombre))? organizaci[oó]n_?pol[ií]tica
movimiento_?pol[ií]tico filiaci[oó]n_?(pol[ií]tica|partidaria) afiliaci[oó]n_?
(pol[ií]tica|partidaria|partido) militancia(_?pol[ií]tica)? militante
(cod|codigo|nombre|nom)_?partido political_?party party_?membership

KNOWN VALUES EC_L3_FILIACION_POLITICA - List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-partidos.txt	25	80%	Revolución Ciudadana, Revolucion Ciudadana, Acción Democrática Nacional, Accion Democratica Nacional, Movimiento CREO, Partido Social Cristiano

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by another organization from the list; a flag stays a flag.

EC_PARTIDO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_PARTIDO_LISTA .

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-si-no.txt , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-true-false.txt , chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-yes-no.txt , chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-s-n.txt , chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-y-n.txt , chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-0-1.txt , chosen by a keyed hash.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EC_PARTIDO_LISTA SECURE LOOKUP

Replaces the value with one of the 13 values in ec-partidos.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Movimiento Suma	→	Avanza
S	→	S

10.9 Trade union membership

EC_L3_AFILIACION_SINDICAL

COLUMN NAME **KNOWN VALUES** **REGEX DECOMPOSE**

WHAT IT IS

The union, the comité de empresa, the membership and the dues deducted on the payroll. The law does not name it, but the open clause of art. 4 reaches it: it is data whose improper processing discriminates.

HOW IT IS FOUND

COLUMN NAME EC_L3_AFILIACION_SINDICAL - Path

95% sindicato(_?(nombre|afiliado|afiliaci[oó]n|codigo))? sindicalizado afiliado_?sindicato
afiliaci[oó]n_?sindical cuota_?sindical descuento_?sindical aporte_?sindical
asociaci[oó]n_?(sindical|de_?trabajadores) comit[eé]_?de_?empresa contrato_?colectivo
fuero_?sindical dirigente_?sindical trade_?union union_?member(ship)?

KNOWN VALUES EC_L3_AFILIACION_SINDICAL - List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-sindicatos.txt	12	60%	FUT, CEOSL, CEDOCUT, CTE, UGTE, UNE

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

A flag stays a flag; any other value becomes Sin información .

EC_CATEGORIA_SUPRIMIDA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_SUPRIMIR .

EC_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-si-no.txt , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-true-false.txt , chosen by a keyed hash.

EC_BANDERA_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-yes-no.txt , chosen by a keyed hash.

EC_BANDERA_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-s-n.txt , chosen by a keyed hash.

EC_BANDERA_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-y-n.txt , chosen by a keyed hash.

EC_BANDERA_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-0-1.txt , chosen by a keyed hash.

EC_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EC_SUPRIMIR **SECURE LOOKUP**

Replaces the value with one of the 1 values in ec-sin-informacion.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
UNE	→	Sin información
S	→	S

10.10 Biometric data

EC_L3_BIOMETRICO

COLUMN NAME **DATA TYPE** **REGEX DECOMPOSE**

WHAT IT IS

Fingerprint templates — the cédula's fingerprint is the country's identification pattern —, face, iris, voice and digitized signature. Article 4 defines the biometric datum as the one that permits or confirms the unique identification of a person.

HOW IT IS FOUND

COLUMN NAME EC_L3_BIOMETRICO - Path

90% biometr[ií]a[a-z]* biom[eé]trico[a-z]* huella(_?(dactilar|digital))?(_(?
(template|plantilla|imagen|hash))?) huellas dactilar dactiloscop[ií]a template_
(facial|huella|biometrico) plantilla_?(facial|biom[eé]trica) rostro_(hash|template)
reconocimiento_?facial firma_?(biom[eé]trica|digitalizada|electr[oó]nica) iris voz_
(biom[eé]trica|template) fingerprints? face_?(template|hash|encoding) biometric[a-z]*

DATA TYPE EC_L3_BIOMETRICO - Type

Accepts text; any other type rules the domain out.

HOW IT IS MASKED

Every character becomes x . A masked template authenticates nobody.

EC_REDACTAR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE		MASKED
A1B2C3D4E5F6	→	XXXXXXXXXXXX

10.11 Genetic data

EC_L3_GENETICO

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Results of genetic and DNA tests, paternity tests, genetic profiles, mutations, karyotype and newborn screening. Article 4 defines the genetic datum as the one that gives unique information on the physiology or health of an individual.

HOW IT IS FOUND

COLUMN NAME EC_L3_GENETICO - Path

90% adn dna gen[eé]tic[oa]s? datos_?gen[eé]ticos? genoma genotip[a-z]* prueba_
(gen[eé]tica|de_?adn|de_?paternidad|de_?filiaci[oó]n) perfil_?gen[eé]tico mutaci[oó]n
brca[12]? cariotipo tamizaje_?neonatal genetic[a-z]*

HOW IT IS MASKED

Every character becomes X .

EC_REDACTAR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE		MASKED
BRCA1 positivo	→	XXXXXXXXXXXX

10.12 Sexual orientation

EC_L3_ORIENTACION_SEXUAL

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Self-declared sexual orientation or preference, and the sexual life of the same sentence of article 4.

HOW IT IS FOUND

COLUMN NAME	EC_L3_ORIENTACION_SEXUAL - Path		
95%	orientaci[o6]n_?sexual	preferencias?_?sexuales?	(cod codigo tipo)_?orientacion_?sexual
	vida_?sexual	sexual_?orientation	
KNOWN VALUES	EC_L3_ORIENTACION_SEXUAL - List		
LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-orientaciones-sexuales.txt	14	80%	Heterosexual, Gay, Lesbiana, Bisexual, Pansexual, Asexual

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by a category from the list.

EC_ORIENTACION_SEXUAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_ORIENTACION_SEXUAL_LISTA .

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-si-no.txt , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-true-false.txt , chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-yes-no.txt , chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-s-n.txt , chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-y-n.txt , chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-0-1.txt , chosen by a keyed hash.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EC_ORIENTACION_SEXUAL_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in ec-orientaciones-sexuales.txt , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Homosexual	→ Pansexual

10.13 Gender identity

EC_L3_IDENTIDAD_GENERO

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

The **identidad de género** article 4 names second in its list, the social name and the intersex condition.

HOW IT IS FOUND

COLUMN NAME

EC_L3_IDENTIDAD_GENERO - Path

95% identidad_?(de_?)?g[eé]nero (cod|codigo|tipo)_?identidad_?genero g[eé]nero_? (autopercibido|identitario) expresi[oó]n_?de_?g[eé]nero transg[eé]nero persona_?trans intersexual nombre_?social pronombres? gender_?identity

KNOWN VALUES

EC_L3_IDENTIDAD_GENERO - List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-identidades-genero.txt	21	70%	Mujer cisgénero, Mujer cisgenero, Hombre cisgénero, Hombre cisgenero, Mujer trans, Hombre trans

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by a category from the list.

EC_IDENTIDAD_GENERO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_IDENTIDAD_GENERO_LISTA .

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-si-no.txt , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-true-false.txt , chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-yes-no.txt , chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-s-n.txt , chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-y-n.txt , chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-0-1.txt , chosen by a keyed hash.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EC_IDENTIDAD_GENERO_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in ec-identidades-genero.txt , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Transgénero	→ Otra

COLUMN NAME	REGEX DECOMPOSE
-------------	-----------------

WHAT IT IS

Being a victim of gender, domestic or sexual violence, of trafficking or of enforced disappearance, the type of violence and the protection measures. The Ley Orgánica Integral para Prevenir y Erradicar la Violencia contra las Mujeres protects the identity of the victim.

HOW IT IS FOUND

COLUMN NAME	EC_L3_VICTIMA - Path
90%	<pre>v[íi]ctima(_?(de_)?(violencia(_?(de_)?g[eé]nero intrafamiliar f[íi]sica sexual psicol[oó]gica econ[oó]mica))? [delito trata abuso desaparici[oó]n))? violencia_?(de_)?g[eé]nero violencia_? (intrafamiliar dom[eé]stica sexual psicol[oó]gica) tipo_?(de_)?violencia medida_?(de_)? protecci[oó]n boleta_de_?auxilio desaparici[oó]n_?forzada trata_?(de_)?personas denunciante_?violencia</pre>

HOW IT IS MASKED

A flag stays a flag; any other value becomes Sin información .

EC_CATEGORIA_SUPRIMIDA	REGEX DECOMPOSE
------------------------	-----------------

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_SUPRIMIR .

EC_BANDERA	REGEX DECOMPOSE
	Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
EC_BANDERA_SI_NO	SECURE LOOKUP Replaces the value with one of the 2 values in ec-bandera-si-no.txt , chosen by a keyed hash.
EC_BANDERA_TRUE_FALSE	SECURE LOOKUP Replaces the value with one of the 2 values in ec-bandera-true-false.txt , chosen by a keyed hash.
EC_BANDERA_YES_NO	SECURE LOOKUP Replaces the value with one of the 2 values in ec-bandera-yes-no.txt , chosen by a keyed hash.
EC_BANDERA_S_N	SECURE LOOKUP Replaces the value with one of the 2 values in ec-bandera-s-n.txt , chosen by a keyed hash.
EC_BANDERA_Y_N	SECURE LOOKUP Replaces the value with one of the 2 values in ec-bandera-y-n.txt , chosen by a keyed hash.
EC_BANDERA_0_1	SECURE LOOKUP Replaces the value with one of the 2 values in ec-bandera-0-1.txt , chosen by a keyed hash.
EC_CM_ALFANUM	CHARACTER MAPPING Maps each character to another of its group (7 groups), keeping length and format.
EC_SUPRIMIR	SECURE LOOKUP Replaces the value with one of the 1 values in ec-sin-informacion.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Violencia intrafamiliar	→	Sin información
S	→	S

Health data

Health data is sensitive by article 4 and a special category by article 25 c). Article 31.2 asks in so many words that it be **previously anonymized or pseudonymized whenever possible**, and article 25 d) adds the data of persons with disabilities and of their substitutes as a category of its own.

11.1 Health identifiers

EC_SALUD_IDENTIFICADOR

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

Clinical record, admission, care, appointment and prescription numbers, and the RDACAA identifier of an outpatient visit.

HOW IT IS FOUND

COLUMN NAME	EC_SALUD_IDENTIFICADOR – Path
85%	<code>n(o ro um umero)?(historia(_?cl[ií]nica)? hc hcu expediente(_?cl[ií]nico)? ingreso atenci[oó]n consulta cita referencia_?m[eé]dica receta) historia_?cl[ií]nica(_?[uú]nica)? hc_?(n(o ro um umero))? hcu n(o ro um umero)?(asegurado afiliado_?iess) c[oó]digo?(paciente historia atenci[oó]n) id_?paciente rdacaa</code>
55%	<code>expediente atenci[oó]n ingreso consulta</code>

HOW IT IS MASKED

Character by character, keeping length and format.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
HC-2024-00123	→ TW-3389-01317

11.2 Health insurance

EC_SALUD_COBERTURA

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

The IESS general insurance, the Seguro Social Campesino, ISSFA, ISSPOL, the public system or a private insurer, by name or code. Which one a person has says a great deal about them.

HOW IT IS FOUND

COLUMN NAME	EC_SALUD_COBERTURA – Path
85%	<code>cobertura(_?(m[eé]dica de_?salud social))? seguro_?(m[eé]dico salud) r[eé]gimen(_?(de_)?salud)? tipo_?(de_)?seguro proveedor_?(de_)?salud iess(_?(afiliado cobertura))? issfa isspol seguro_?campesino msp aseguradora_?(m[eé]dica salud) plan_?(m[eé]dico de_?salud) medicina_?prepagada</code>

KNOWN VALUES EC_SALUD_COBERTURA – List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-cobertura.txt	19	80%	IESS (Seguro General), Seguro Social Campesino, ISSFA, ISSPOL, Ministerio de Salud Pública, Ministerio de Salud Publica

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A name is replaced by another cover from the list; a code is mapped digit by digit; a flag stays a flag.

EC_COBERTURA_SALUD REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_COBERTURA .

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-si-no.txt , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-true-false.txt , chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-yes-no.txt , chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-s-n.txt , chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-y-n.txt , chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-0-1.txt , chosen by a keyed hash.

EC_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EC_COBERTURA SECURE LOOKUP

Replaces the value with one of the 8 values in ec-cobertura-salud.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
IESS	→	ISSPOL
12345	→	31564

11.3 Diagnosis and ICD-10

EC_SALUD_DIAGNOSTICO

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

Diagnoses in words or by ICD-10 code, with a dot (F33.1) or without (F331); causes of death, of sick leave and of admission; allergies and comorbidities.

HOW IT IS FOUND

COLUMN NAME EC_SALUD_DIAGNOSTICO – Path

85% `diagn[oó]stico[a-z]*   dx(?:principal|secundario|ingreso|egreso)[0-9]?   cie_?10   cie(?:`
`(10|11|principal))?)?   (cod|codigo)_?(diagnostico|dx|cie)   impresi[oó]n_?diagn[oó]stica`
`enfermedad(es)?[a-z]*   patolog[ií]a   comorbilidad(es)?   antecedentes_?`
`(m[eé]dicos|patol[oó]gicos|cl[ií]nicos|personales)   causa_?(de_)?`
`(muerte|defunci[oó]n|incapacidad|hospitalizaci[oó]n|consulta)   motivo_?(de_)?`
`(consulta|reposo|hospitalizaci[oó]n)   alergias?   diagnos(is|es)   icd_?(10|11)?`

VALUE FORMAT EC_SALUD_DIAGNOSTICO – Regex

PATTERN	STRENGTH	NOTES
<code>[A-TV-Za-tv-z]\d{2}(\.\d{1,2} \d X)?</code>		50%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

An ICD-10 code becomes another common code in the same format, and text a common diagnosis: an HIV or mental disorder diagnosis does not survive as such.

EC_DIAGNOSTICO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches:

EC_DIAGNOSTICO_TEXT0 .

EC_CIE10_DECIMAL SECURE LOOKUP

Replaces the value with one of the 33 values in `ec-cie10-decimal.txt` , chosen by a keyed hash.

EC_CIE10 SECURE LOOKUP

Replaces the value with one of the 37 values in `ec-cie10.txt` , chosen by a keyed hash.

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt` , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt` , chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt` , chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt` , chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.

EC_DIAGNOSTICO_TEXT0 SECURE LOOKUP

Replaces the value with one of the 30 values in `ec-diagnostics.txt` , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
F33.1	→	K21.9
B24X	→	R51

11.4 Procedures

EC_SALUD_PROCEDIMIENTO

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Procedures, tests and surgeries, in words or by code. A viral load or a pregnancy test reveals the diagnosis.

HOW IT IS FOUND

COLUMN NAME EC_SALUD_PROCEDIMIENTO - Path

85% procedimiento(?:([m[eé]dico|quir[uú]rgico|realizado|autorizado)))? (cod|codigo)?
(procedimiento|prestaci[oó]n|servicio_?salud|cpt) cpt prestaci[oó]n(?:([m[eé]dica|de_?salud)
examen?(ordenado|realizado) cirug[ií]a(?:realizada)? intervenci[oó]n_?quir[uú]rgica
medical_?procedure

HOW IT IS MASKED

A code is mapped digit by digit; text is replaced by a common procedure.

EC_PROCEDIMIENTO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches:

EC_PROCEDIMIENTO_TEXTO .

EC_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EC_PROCEDIMIENTO_TEXTO SECURE LOOKUP

Replaces the value with one of the 10 values in ec-procedimientos.txt , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
901234	→ 928120

11.5 Medicines

EC_SALUD_MEDICAMENTO

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Prescribed or dispensed medicines, by name or active ingredient. An antiretroviral, an antipsychotic or a contraceptive says what the diagnosis column was masked for.

HOW IT IS FOUND

COLUMN NAME EC_SALUD_MEDICAMENTO - Path

85% medicamentos(?:([recetado|prescrito|entregado|despachado|nombre|uso]))? f[aá]rmacos?
principio_?activo denominaci[oó]n_?com[uú]n_?internacional dci receta(?:[m[eé]dica)?
prescripci[oó]n(?:medicamento)? posolog[ií]a tratamiento_?farmacol[oó]gico (cod|codigo)?
medicamento cuadro_?nacional(?:de_?)?medicamentos medications? drugs_?prescribed

KNOWN VALUES EC_SALUD_MEDICAMENTO - List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar- medicamentos.txt	50	70%	Paracetamol, Ibuprofeno, Diclofenaco, Metformina, Glibenclamida, Losartán

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Replaced by a common medicine from the list; a flag stays a flag.

EC_MEDICAMENTO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_MEDICAMENTO_LISTA

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EC_MEDICAMENTO_LISTA SECURE LOOKUP

Replaces the value with one of the 22 values in `ec-medicamentos.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Sertralina 50 mg

→ **Complejo B**

11.6 Clinical text

EC_SALUD_TEXTO_CLINICO

COLUMN NAME

DATA TYPE

FREE TEXT REDACTION

WHAT IT IS

History, present illness, progress notes, physical examination, treatment plan, discharge summaries and nursing notes.

HOW IT IS FOUND

COLUMN NAME

EC_SALUD_TEXTO_CLINICO - Path

85%

```
anamnesis  evoluci[oó]n(_?(m[eé]dica|cl[ií]nica|enfermer[ií]a))?  enfermedad_?actual
examen_?f[ií]sico  plan_?(de_)?(manejo|tratamiento)  indicaciones_?m[eé]dicas  conducta(_?
m[eé]dica)?  epicrisis  resumen_?(de_)?(alta|egreso|atenci[oó]n)  nota_?(m[eé]dica|de_?
enfermer[ií]a|cl[ií]nica)  notas?_?(m[eé]dicas|cl[ií]nicas)  triaje(_?texto)?  informe_?
(patolog[ií]a|radiolog[ií]a|m[eé]dico)  clinical_?notes?  discharge_?summary
```

DATA TYPE

EC_SALUD_TEXTO_CLINICO - Type

Accepts text (length 30 or more); any other type rules the domain out.

HOW IT IS MASKED

The same Free Text Redaction as free text: documents, contacts and names from the list are redacted. The clinical content stays, because that is what makes the copy useful.

EC_TEXTO_LIBRE FREE TEXT REDACTION

Replaces the text matching 8 patterns with [DATO], and the 1,668 terms of ec-texto-nombres.txt with [NOMBRE].

EXAMPLES

VALUE

Paciente Zambrano, cédula 1710034065,
consulta por cefalea.

MASKED

→ Paciente [NOMBRE], cédula [DATO], consulta
por cefalea.

11.7 Test results

EC_SALUD_RESULTADO_EXAMEN

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

Laboratory and test results — glucose, pregnancy, toxicology, blood alcohol —, also as numeric values.

HOW IT IS FOUND

COLUMN NAME

EC_SALUD_RESULTADO_EXAMEN – Path

80%

resultado_(del_)?

(examen|prueba|laboratorio|pcr|serolog[ií]a|biopsia|glucosa|citolog[ií]a|papanicolaou)

(examen|prueba)_(resultado|laboratorio) glucemia hemoglobina(_?glicosilada)? hba1c

colesterol imc presi[oó]n_arterial prueba_(de_)?embarazo toxicol[oó]gico

alcoholemia lab_results? test_results?

50%

resultado prueba

HOW IT IS MASKED

Character by character, keeping the format: a number stays a number.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

Reactivo

MASKED

→ Quilqaju

11.8 HIV and sexually transmitted infections

EC_SALUD_VIH

COLUMN NAME

REGEX DECOMPOSE

WHAT IT IS

Serological status, viral load and antiretroviral treatment, hepatitis, tuberculosis and sexually transmitted infections. The Ley de Prevención y Atención Integral del VIH/SIDA orders confidentiality and forbids demanding a test to get or keep a job.

HOW IT IS FOUND

COLUMN NAME EC_SALUD_VIH – Path

90% vih(_?(estado|resultado|prueba|diagn[ó]stico|positivo))? hiv sida aids serolog[ií]a?
(vih|hiv) estado?serol[ó]gico carga?viral cd4 tarv antirretroviral(es)? its ets
infecci[ó]n(es)?_(de_)?transmisi[ó]n?sexual s[ií]filis hepatitis?[bc] tuberculosis
tb

HOW IT IS MASKED

A flag stays a flag; any other value becomes Sin información .

EC_CATEGORIA_SUPRIMIDA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_SUPRIMIR .

EC_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-si-no.txt , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-true-false.txt , chosen by a keyed hash.

EC_BANDERA_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-yes-no.txt , chosen by a keyed hash.

EC_BANDERA_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-s-n.txt , chosen by a keyed hash.

EC_BANDERA_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-y-n.txt , chosen by a keyed hash.

EC_BANDERA_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in ec-bandera-0-1.txt , chosen by a keyed hash.

EC_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EC_SUPRIMIR **SECURE LOOKUP**

Replaces the value with one of the 1 values in ec-sin-informacion.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Reactivo	→	Sin información
S	→	S

11.9 Blood group and Rh

EC_SALUD_GRUPO_SANGUINEO

COLUMN NAME **KNOWN VALUES** **SECURE LOOKUP**

WHAT IT IS

Blood group and Rh factor, which the cédula itself carries.

HOW IT IS FOUND

COLUMN NAME EC_SALUD_GRUPO_SANGUINEO – Path

90% grupo?sangu[ií]neo tipo?(de_)?sangre rh factor?rh gs?rh grupo?rh abo(_?rh)?
blood?(type|group)

KNOWN VALUES EC_SALUD_GRUPO_SANGUINEO - List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-grupos-sanguineos.txt	18	90%	A+, A-, B+, B-, AB+, AB-

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by another group from the list.

EC_GRUPO_SANGUINEO **SECURE LOOKUP**

Replaces the value with one of the 8 values in `ec-grupos-sanguineos.txt`, chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
0+	→ A-

11.10 Disability

EC_SALUD_DISCAPACIDAD

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Disability status, type and percentage, the CONADIS card and the substitute. Article 25 d) makes the data of persons with disabilities **and of their substitutes** a special category of its own – no other law of the region does.

HOW IT IS FOUND

COLUMN NAME EC_SALUD_DISCAPACIDAD - Path

90% `discapacidad[a-z_]* (tipo|cod|codigo|categoria|porcentaje|grado)?discapacidad persona_? con_?discapacidad pcd conadis carn[eé]_?(de_?)(conadis|discapacidad) certificado_? (de_?)discapacidad condici[oó]n_?(de_?)discapacidad sustituto(_?de_?persona_?con_? discapacidad)? necesidades_?(educativas_?)especiales movilidad_reducida disabilit(y|ies)`

KNOWN VALUES EC_SALUD_DISCAPACIDAD - List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-discapacidades.txt	27	60%	Física, Fisica, Auditiva, Visual, Intelectual, Psicosocial

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced by a category from the list.

EC_DISCAPACIDAD **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_DISCAPACIDAD_LISTA .

EC_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

EC_BANDERA_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera=yes-no.txt` , chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt` , chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EC_DISCAPACIDAD_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `ec-discapacidades.txt` , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Trastorno del espectro autista

→

Ninguna

11.11 Occupational health

EC_SALUD_OCUPACIONAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The fitness opinion of occupational medical exams, work accidents and occupational diseases, sick leave and absenteeism.

HOW IT IS FOUND

COLUMN NAME

EC_SALUD_OCUPACIONAL - Path

85%

```
aptitud_?(laboral|m[é]dica) certificado_?(de_)?aptitud examen_?(m[é]dico_)?  
(ocupacional|preocupacional|de_ingreso|de_retiro|peri[ód]ico) accidente_?(de_)?trabajo  
enfermedad_?(laboral|profesional|ocupacional) riesgos_?(del_)?trabajo incapacidad(_?  
(m[é]dica|laboral|d[ií]as))? reposo_?m[é]dico d[ií]as_?(de_)?(incapacidad|reposo)  
licencia_?(m[é]dica|por_?enfermedad|de_maternidad) ausentismo comit[é]_?(de_)?  
seguridad
```

KNOWN VALUES

EC_SALUD_OCUPACIONAL - List

LIST

VALUES

STRENGTH

FIRST VALUES

ec-detectar-aptitud-
laboral.txt

5

60%

apto, no apto, apto con restricciones, pendiente, apto con
recomendaciones

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

An opinion is replaced by another from the list — fit, unfit —; a flag stays a flag; the diagnosis falls under its own domain.

EC_APTITUD_LABORAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_APTITUD_LABORAL_LISTA .

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt` , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt` , chosen by a keyed hash.

EC_BANDERA_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt` , chosen by a keyed hash.

EC_BANDERA_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.

EC_BANDERA_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt` , chosen by a keyed hash.

EC_BANDERA_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.

EC_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EC_APTITUD_LABORAL_LISTA **SECURE LOOKUP**

Replaces the value with one of the 4 values in `ec-aptitud-laboral.txt` , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

No apto temporalmente

→ **Pendiente**

11.12 Sexual and reproductive health

EC_SALUD_REPRODUCTIVA

COLUMN NAME

REGEX DECOMPOSE

WHAT IT IS

Pregnancy, last menstrual period, prenatal care, childbirth, family planning, fertility.

HOW IT IS FOUND

COLUMN NAME

EC_SALUD_REPRODUCTIVA - Path

90%

embarazo embarazada gestante gestaci[oó]n edad_?gestacional semanas_?(de_)?
gestaci[oó]n fum fecha_?[uú]ltima_?(regla|menstruaci[oó]n) control_?prenatal prenatal
parto(_?tipo)? ces[aá]rea aborto m[eé]todo_?(de_)?planificaci[oó]n planificaci[oó]n_
familiar anticoncepci[oó]n anticonceptivo fertilidad salud_?(sexual|reproductiva)
pregnan(t|cy) contracepti(on|ve)

HOW IT IS MASKED

Flags keep their shape; any other value becomes Sin información .

EC_CATEGORIA_SUPRIMIDA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_SUPRIMIR .

EC_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt` , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt` , chosen by a keyed hash.

EC_BANDERA_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt` , chosen by a keyed hash.

EC_BANDERA_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.

EC_BANDERA_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt` , chosen by a keyed hash.

EC_BANDERA_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.

EC_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EC_SUPRIMIR **SECURE LOOKUP**

Replaces the value with one of the 1 values in `ec-sin-informacion.txt` , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
12 semanas de gestación	→	Sin información
S	→	S

11.13 Mental health

EC_SALUD_MENTAL

COLUMN NAME

REGEX DECOMPOSE

WHAT IT IS

Mental disorders, psychiatric or psychological care, suicide attempts and substance use. Article 4 defines *datos relativos a la salud* as covering physical **and mental** health.

HOW IT IS FOUND

COLUMN NAME

EC_SALUD_MENTAL – Path

90%

```
salud_?mental  trastorno[a-z]*  psiqui[aá]tri[a-z]*  psicol[oó]gi[a-z]*  depresi[oó]n
ansiedad  suicid[a-z]*  intento_?(de_)?suicidio  autolesi[oó]n  consumo_?(de_)?
(sustancias|alcohol|drogas)  adicci[oó]n(es)?  alcoholismo  tabaquismo  mental_?health
```

HOW IT IS MASKED

Flags keep their shape; any other value becomes `Sin información` .

EC_CATEGORIA_SUPRIMIDA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_SUPRIMIR` .

EC_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt` , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt` , chosen by a keyed hash.

EC_BANDERA_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt` , chosen by a keyed hash.

EC_BANDERA_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.

EC_BANDERA_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt` , chosen by a keyed hash.

EC_BANDERA_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.

EC_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EC_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in `ec-sin-informacion.txt` , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Trastorno bipolar

→

Sin información

Credit and socioeconomic data

Article 4 defines the **datos personales crediticios** as those that make up the economic behaviour of a natural person, and articles 28 and 29 give them a regime of their own: only to analyse solvency, never for a secondary purpose, and not communicated once five years have passed. The rest of the group is not credit data, but it identifies: a debt, an asset, a pension, a social programme or a Registro Social quintile narrows a person down as fast as a postcode.

12.1 Credit history

EC_FIN_HISTORIAL_CREDITICIO

COLUMN NAME

REGEX DECOMPOSE

WHAT IT IS

The bureau report, the score, days in arrears and the risk category the Superintendencia de Bancos uses — A-1 to E. Article 4 defines these as *datos personales crediticios*, and article 28 allows them only for analysing solvency, with **no secondary purpose** and no communication after five years.

HOW IT IS FOUND

COLUMN NAME

EC_FIN_HISTORIAL_CREDITICIO – Path

90%

```
historial_?(crediticio|de_?cr[eé]dito) bur[oó]_?(de_?)?cr[eé]dito central_?(de_?)?
riesgos? equifax credit_?report score(_?(crediticio|interno))? puntaje_?(de_?)?
cr[eé]dito calificaci[oó]n_?(de_?)?(riesgo|cr[eé]dito|cartera|deudor) categor[ií]a_?(de_?)?
riesgo d[ií]as_?(de_?)?(mora|atraso) estado_?(de_?)?(cr[eé]dito|cartera|obligaci[oó]n)
morosidad moroso cobranza_?judicial credit_?score credit_?rating
```

HOW IT IS MASKED

A score gets other digits of the same length; a flag stays a flag; a category is replaced by another from the list.

EC_HISTORIAL_CREDITICIO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_ESTADO_CREDITO .

EC_CM_DIGITOS

CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EC_BANDERA

REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

EC_ESTADO_CREDITO SECURE LOOKUP

Replaces the value with one of the 15 values in `ec-estados-credito.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Reportado en el buró de crédito	→	Refinanciado
745	→	471

12.2 Debt

EC_FIN_DEUDA

COLUMN NAME DATA TYPE NUMERIC MAPPING

WHAT IT IS

Credit and portfolio balances, amounts in arrears, instalments, disbursed amounts and limits.

HOW IT IS FOUND

COLUMN NAME EC_FIN_DEUDA – Path

85% saldo_?(deuda|cr[eé]dito|capital|obligaci[oó]n|cartera|en_?mora|adeudado) monto_?(de_?la_?)?(deuda|cuota|obligaci[oó]n|cr[eé]dito|pr[eé]stamo|mora|desembolsado) deuda(_?total)? l[ií]mite_?(de_?)?(cr[eé]dito|tarjeta) cupo_?(de_?)?cr[eé]dito cuota_?mensual loan_?(amount|balance) outstanding_?balance debt

DATA TYPE EC_FIN_DEUDA – Type

Accepts number; any other type rules the domain out.

HOW IT IS MASKED

Mapped to a value between 100 and 250,000 dollars, unrelated to the real one – Ecuador uses the US dollar.

EC_VALOR_DEUDA NUMERIC MAPPING

Maps the number to another between 100 and 250,000.

EXAMPLES

VALUE		MASKED
8400	→	5981

12.3 Income and pay

EC_FIN_INGRESOS

COLUMN NAME DATA TYPE NUMERIC MAPPING

WHAT IT IS

Salary, monthly or household income, fees and rents. The unified basic wage was 470 dollars a month in 2025.

HOW IT IS FOUND

COLUMN NAME EC_FIN_INGRESOS – Path

85% sueldo(_?(b[aá]sico|mensual|bruto|neto|unificado))? salario(_?(b[aá]sico|mensual|bruto|neto))? remuneraci[oó]n(_?(b[aá]sica|mensual|unificada))? rmu ingresos?(_?(mensuales?|totales?|familiares?|declarados?|netos?))? honorarios renta(_?(bruta|neta|mensual|anual|imponible))? ingreso_?(base|familiar|percapita) salary income wages?

DATA TYPE EC_FIN_INGRESOS – Type

Accepts number; any other type rules the domain out.

HOW IT IS MASKED

Mapped to a value between 450 and 9,000 dollars.

EC_INGRESOS **NUMERIC MAPPING**

Maps the number to another between 450 and 9,000.

EXAMPLES

VALUE		MASKED
1200	→	1191

12.4 Assets

EC_FIN_PATRIMONIO

COLUMN NAME **DATA TYPE** **NUMERIC MAPPING**

WHAT IT IS

Net worth, assets, valuations and cadastral values, account and fund balances, term deposits and the sworn declaration of assets.

HOW IT IS FOUND

COLUMN NAME EC_FIN_PATRIMONIO – Path

85% patrimonio(_?(neto|líquido))? activos?_(totales)? valor?_(del_)?
(inmueble|predio|vehículo|bienes|aval[uó]o|comercial) aval[uó]o(_?catastral)? saldo_
(cuenta|ahorros?|depósito|inversión|fondo) depósito?a_plazo inversiones
declaración?(juramentada|patrimonial) net_worth account_balance

DATA TYPE EC_FIN_PATRIMONIO – Type

Accepts number; any other type rules the domain out.

HOW IT IS MASKED

Mapped to a value between 1,000 and 600,000 dollars.

EC_PATRIMONIO **NUMERIC MAPPING**

Maps the number to another between 1,000 and 600,000.

EXAMPLES

VALUE		MASKED
85000	→	78007

12.5 Pension and benefits

EC_FIN_PENSION

COLUMN NAME **DATA TYPE** **NUMERIC MAPPING**

WHAT IT IS

The amount of the pension, subsidy or benefit, and child support.

HOW IT IS FOUND

COLUMN NAME EC_FIN_PENSION – Path

90% pensi[oó]n(_?(monto|mensual|valor|jubilador))? monto_?
(pensi[oó]n|jubilaci[oó]n|beneficio|subsidiario) valor_?(pensi[oó]n|subsidiario|beneficio)
jubilaci[oó]n(_?monto)? pensi[oó]n_(de_)?alimentos pensi[oó]n_alimenticia sut[eé]
pension_amount

DATA TYPE EC_FIN_PENSION – Type

Accepts number; any other type rules the domain out.

HOW IT IS MASKED

Mapped to a value between 200 and 3,000 dollars.

EC_VALOR_PENSION **NUMERIC MAPPING**

Maps the number to another between 200 and 3,000.

EXAMPLES

VALUE	MASKED
520	→ 528

12.6 Social programmes

EC_FIN_PROGRAMA_SOCIAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The Bono de Desarrollo Humano, the Bono Joaquín Gallegos Lara, the pensions for older adults and for persons with disabilities, the Crédito de Desarrollo Humano, by name or as a flag. Being in one says the household is poor.

HOW IT IS FOUND

COLUMN NAME EC_FIN_PROGRAMA_SOCIAL – Path

90% programa_social beneficiario_(programa|subsidiario|bono|pensi[oó]n) bono(_?(desarrollo_?
humano|joaqu[í]n|gallegos|1000_d[ií]as|contingencia))? bdh pensi[oó]n_(adultos?_
mayores|toda_una_vida) cr[eéd]ito_(de_)?desarrollo_humano registro_social
subsidiario_(gas|electricidad|tipo)? transferencia_monetaria ayuda_(social|estatal)
social_programme

KNOWN VALUES EC_FIN_PROGRAMA_SOCIAL – List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-programas-sociales.txt	19	80%	Bono de Desarrollo Humano, Bono Joaquín Gallegos Lara, Bono Joaquin Gallegos Lara, Pensión para Adultos Mayores, Pension para Adultos Mayores, Pensión para Personas con Discapacidad
ec-detectar-codigos-banderas.txt	11	30%	s, n, sí, si, no, 1

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced by a programme from the list.

EC_PROGRAMA_SOCIAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_PROGRAMA_SOCIAL_LISTA

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EC_PROGRAMA_SOCIAL_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in `ec-programas-sociales.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Bono de contingencia

→ **Pensión para Personas con Discapacidad**

12.7 Registro Social

EC_FIN_REGISTRO_SOCIAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The score and quintile the Registro Social gives every household, which decides who gets a subsidy, and the socioeconomic level.

HOW IT IS FOUND

COLUMN NAME

EC_FIN_REGISTRO_SOCIAL - Path

90%

`registro_?social [ii]ndice_?(de_)?registro_?social puntaje_?(registro_?social|rs)`
`clasificaci[oó]n_?socioecon[oó]mica nivel_?socioecon[oó]mico nse condici[oó]n_?(de_)?`
`pobreza pobreza quintil(_?(de_)?(ingreso|riqueza))? decil estrato_?socioecon[oó]mico`

KNOWN VALUES

EC_FIN_REGISTRO_SOCIAL - List

LIST

VALUES

STRENGTH

FIRST VALUES

`ec-detectar-registro-`
`social.txt`

19

60%

Quintil 1, Quintil 2, Quintil 3, Quintil 4, Quintil 5, Pobreza extrema

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A flag stays a flag, a code is replaced by another code and any other value by a class from the list.

EC_REGISTRO_SOCIAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_REGISTRO_SOCIAL_LISTA .

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-si-no.txt , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-true-false.txt , chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-yes-no.txt , chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-s-n.txt , chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-y-n.txt , chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-0-1.txt , chosen by a keyed hash.

EC_REGISTRO_SOCIAL_CODIGO SECURE LOOKUP

Replaces the value with one of the 5 values in ec-registro-social-codigos.txt , chosen by a keyed hash.

EC_REGISTRO_SOCIAL_LISTA SECURE LOOKUP

Replaces the value with one of the 9 values in ec-registro-social.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Pobreza extrema	→	Quintil 1
2	→	4

12.8 Housing tenure

EC_FIN_VIVIENDA

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Housing tenure in the categories the census uses: owned outright or being paid for, donated or inherited, lent, by services, rented, or held in anticresis.

HOW IT IS FOUND

COLUMN NAME EC_FIN_VIVIENDA - Path

90% tenencia?(de_)?(la_)?vivienda tipo?(de_)?tenencia vivienda?(propia|arrendada|tipo_?tenencia) condici[oó]n?(de_)?(la_)?vivienda ocupaci[oó]n?vivienda anticresis housing_tenure

KNOWN VALUES EC_FIN_VIVIENDA - List

LIST	VALUES	STRENGTH	FIRST VALUES
ec-detectar-tenencia-vivienda.txt	15	60%	Propia y totalmente pagada, Propia y la está pagando, Propia y la esta pagando, Propia (regalada, donada, heredada), Prestada o cedida, Por servicios

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Replaced by another tenure from the list.

EC_TENENCIA_VIVIENDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_TENENCIA_VIVIENDA_LISTA .

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt` , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt` , chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt` , chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt` , chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EC_TENENCIA_VIVIENDA_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in `ec-tenencia-vivienda.txt` , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Ocupación de hecho

→

Anticresis

Judicial record

Ecuador is the only country of the region whose law names the **pasado judicial** among the sensitive data of article 4. The group is separate only because the shapes are: everything in it is sensitive data, and article 26 forbids processing it unless one of the grounds it lists applies.

13.1 Judicial record

EC_PENAL_ANTECEDENTES

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

The **pasado judicial** article 4 lists among the sensitive data: the record certificate, the type of offence, the procedural situation, the arrest warrant, being deprived of liberty and the SNAI record.

HOW IT IS FOUND

COLUMN NAME EC_PENAL_ANTECEDENTES – Path

95% antecedentes?(? (penales|policiales|judiciales))? pasado_?judicial certificado_?(de_)?
 (antecedentes|pasado_?judicial) r[eé]cord_?policial reincidencia condena(_?
 (penal|tipo))? delito(_?(tipo|cometido))? tipo_?(de_)?delito situaci[oó]n_?jur[ií]dica
 boleta_?(de_)?(captura|excarcelaci[oó]n) privad[oa]_?(de_)?libertad ppl snai centro_?
 (de_)?rehabilitaci[oó]n_?social criminal_?record

55% estado_?(del_)?proceso sentencia fallo

HOW IT IS MASKED

Replaced by a generic status from the list — no record, sentenced, acquitted —, whatever the outcome; a flag stays a flag.

EC_ANTECEDENTES REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_ANTECEDENTES_LISTA .

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt` , chosen by a keyed hash.

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt` , chosen by a keyed hash.

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt` , chosen by a keyed hash.

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt` , chosen by a keyed hash.

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EC_ANTECEDENTES_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in `ec-antecedentes.txt` , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Sentenciado por hurto en 2019	→	Absuelto

13.2 Court file

EC_PENAL_EXPEDIENTE

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The number the Función Judicial gives a case — judicial unit, year and correlative, 17294-2024-00123 — and the file of the Fiscalía.

HOW IT IS FOUND

COLUMN NAME	EC_PENAL_EXPEDIENTE – Path
90%	n(o ro um umero)?(de_)?(expediente?(judicial penal fiscal) causa proceso?(judicial penal) denuncia noticia?del?delito referencia?(judicial fiscal)) expediente?(judicial penal fiscal) n(o ro um umero)?(de_)?juicio juicio?n(o ro um umero) causa? penal noticia?del?delito satje
50%	expediente causa proceso juicio

VALUE FORMAT	EC_PENAL_EXPEDIENTE – Regex	
PATTERN	STRENGTH	NOTES
\d{5}-\d{4}-\d{5}		90%
\d{1,6}-[A-Z]{2,10}-\d{4}		50%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The judicial unit and the correlative are mapped digit by digit; the year stays. Other numbers are masked character by character.

EC_EXPEDIENTE_JUDICIAL REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: EC_CM_ALFANUM .

EC_CM_DIGITOS	CHARACTER MAPPING
Maps each character to another of its group (1 groups), keeping length and format.	
EC_CM_ALFANUM	CHARACTER MAPPING
Maps each character to another of its group (7 groups), keeping length and format.	

EXAMPLES

VALUE		MASKED
17294-2024-00123	→	23323-2024-47916
1234/2023	→	9707/2023

The cédula and RUC check digits

The **cédula** is ten digits. Each of the first nine is multiplied by the series 2 1 2 1 2 1 2 1 2, every product over nine is reduced by nine, and the tenth digit is ten minus the remainder of the sum modulo ten. That is the **Luhn algorithm**, the same one a payment card uses — which is what makes the masking a single step.

The Masking Engine's **Payment Card** framework keeps the Luhn residue of the value it is given: a valid cédula comes out as another valid cédula. Its `preserve` parameter fixes how many leading digits stay, and three of them keep the **province of issue** and the digit that says the holder is a natural person. The province stays for the same reason the provincia column stays: the set generalizes below the province, not at it.

The **RUC** is thirteen digits, and its third one decides the rule:

- **0 to 5 — a natural person:** the RUC is the cédula plus the establishment code `001`. The set masks the ten digits as a cédula and keeps the code, so the cédula and the RUC of the same person still agree.
- **9 — a private company:** the check digit is in the tenth position, computed modulo 11 with the coefficients 4 3 2 7 6 5 4 3 2. **6 — a public body:** the check digit is in the ninth, with the coefficients 3 2 7 6 5 4 3 2. Neither is a natural person, so neither is personal data: the set leaves both exactly as they are, and says so.

Where people live: parroquias, DPA code and postal code

With the population the 2022 census counted and the location the DPA layer publishes, the set knows the size, cantón, provincia and position of about 1,040 parroquias and 224 cantones.

- A **parroquia** with fewer than 20,000 inhabitants becomes the nearest parroquia of its **cantón** with at least 20,000; when its cantón has none, the nearest of its provincia. There are 913 of them, with 4.3 million inhabitants.
- The **93 cantones** under 20,000 follow the same rule inside their provincia. The 24 provincias stay.
- Three provincias have no parroquia of 20,000 at all — Galápagos, Zamora Chinchipe and the zona no delimitada. There the value stays inside the provincia and becomes its most populous parroquia: a smaller group than elsewhere, and the documentation says so rather than sending a Galápagos resident to the mainland.
- Distance is measured between the centroids of the DPA layer, in a straight line.
- Names repeat — San José, Santa Rosa, San Vicente, Bellavista. A name is generalized only when every place bearing it is small, and then follows the most populous of them; when that bearer is a cantón, the replacement is a cantón. Written with the cantón or the provincia (`Facundo Vela - Bolívar`) the same rule applies inside it.
- The **DPA code** follows the same table, so a person's parroquia name and code still agree — and because the postal code is the DPA code, the **ciudadela** is replaced by another and the **postal code** keeps only the provincia.

Parroquia names that are also given names or surnames, and provincia names, are left out of the detection list, so that a surname column is not taken for parroquias.

Limits

- **Binary columns** — cédula photos, fingerprints, validation selfies, clinical PDFs in BLOBs — cannot be masked by these algorithms. Inventory them and report them as residual risk.
- **The province of issue of the cédula stays**, and so does the provincia: the set generalizes below the province, not at it. If the province itself is too revealing for your data, add a step that maps it — remembering that any change to those two digits breaks the Luhn digit, which then has to be recomputed.
- **A company RUC is not masked**: article 4 makes only a natural person a data subject. If your rule set needs it masked anyway, give the column a domain of its own.
- **The DPA layer publishes the place names without their acute accents**. The set restores them with a dictionary, and writes both spellings into the tables, but a name it could not restore comes out unaccented.
- **Data Cleansing leaves unknown values as they are**: a parroquia spelled in a way that is not in the table — abbreviated, misspelled — passes unchanged.
- **Free text** is redacted word by word: a phone written with spaces inside a sentence may not be found, and a name that is also a common word (Rosa , Cruz , Flores) is redacted wherever it appears.
- **Children and adolescents** are a special category of their own (art. 25 b). A birth date moves at most 60 days, so few people cross their 18th birthday — but some do. Where that line matters, split the rule by age group with Multi-Column Condition.
- **Addresses** are replaced as a whole line. Street, ciudadela, parroquia and postal code in separate columns are masked independently and no longer form a coherent address.
- **The engine key** links masked values back to production for whoever holds both: the copy is **pseudonymized** in the sense of article 4, not anonymized, and remains personal data. Keep the key apart and under technical and organizational measures.

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Verification

Before publication, the set was checked outside a Masking Engine:

- The cédula rule was checked against published valid numbers, and the two RUC rules against RUCs published by companies and public bodies.
- Every classifier configuration was reviewed against what a Masking Engine accepts.
- Profiling was simulated on 142 columns with the whole set at its assignment threshold — well named, badly named and 29 columns that are not personal data — and again with only the classifiers of the essential pack.
- The table that generalizes places was audited for all 1,040 parroquias and 224 cantones: by name, by name with the provincia, and by DPA code.
- The algorithms, run with the Delphix algorithm plugin, masked each domain's sample values and placeholders such as N/A , checking every output against its vocabulary; 200 cédulas for validity, distinctness and the province and type they keep, 25 cédula and RUC pairs to confirm they still agree, and 40 company RUCs to confirm they come back untouched. Every algorithm also masked its own sample value.

Every column was assigned as expected and every value masked as expected. Before using the set on real data, run a profiling job and a masking job on a representative copy.

Sources

- Ley Orgánica de Protección de Datos Personales (Fifth Supplement of Official Register 459, 26 May 2021) and its Reglamento General (Official Register Supplement 435, 13 November 2023).
- Constitution of the Republic (indigenous nationalities and pueblos, union of fact). Ley Orgánica de Salud. Ley de Prevención y Atención Integral del VIH/SIDA. Ley Orgánica de Discapacidades. Ley Orgánica Integral para Prevenir y Erradicar la Violencia contra las Mujeres. Código Orgánico Integral Penal.
- Instituto Nacional de Estadística y Censos: VIII Censo de Población y VII de Vivienda 2022, population by parroquia, and the División Política Administrativa with the code and location of every parroquia and cantón.
- Registro Civil (cédula and its check digit). SRI (RUC and its two rules). Superintendencia de Bancos (risk categories A-1 to E). IESS, ISSFA and ISSPOL. CONADIS. MIES (Registro Social and the social programmes). CNE (register of political organizations). Ministry of Labour (unified basic wage 2025). ANT (Placa). Correos del Ecuador (six-digit postal code). SENESCYT.

APPENDIX A

Algorithms

Every algorithm of the set in alphabetical order, with its configuration. The one a domain names is the top of its tree; the others are building blocks.

EC_ANIO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: EC_L2_ANIO_NACIMIENTO

PATTERNS, TRIED IN ORDER

(\d{3})(\d)	keep · EC_CM_DIGITOS
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

EC_ANTECEDENTES REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_ANTECEDENTES_LISTA .

Used by: EC_PENAL_ANTECEDENTES

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,6})	EC_CM_ALFANUM
otherwise	EC_ANTECEDENTES_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

EC_ANTECEDENTES_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in ec-antecedentes.txt , chosen by a keyed hash.

Used by: EC_ANTECEDENTES

CONFIGURATION

lookupFile	ec-antecedentes.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_APELLIDO NAME

Replaces the name with one of the 366 names in ec-apellidos.txt .

Used by: EC_PALABRA_APELLIDO

CONFIGURATION

lookupFile	ec-apellidos.txt
maskedValueCase	PRESERVE_INPUT
filterAccent	yes
inputCaseSensitive	no

EC_APELLIDOS REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: EC_CM_ALFANUM .

Used by: EC_L1_APELLIDO , EC_NOMBRE_COMPLETO

PATTERNS, TRIED IN ORDER

(\S+)	EC_PALABRA_APELLIDO
(\S+)\s+(\S+)	EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO
(\S+)\s+(\S+)\s+(\S+)	EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO
(\S+)\s+(\S+)\s+(\S+)\s+(\S+)	EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO
otherwise	EC_CM_ALFANUM

CONFIGURATION

trimInput	yes
requireMask	no

EC_APTITUD_LABORAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_APTITUD_LABORAL_LISTA .

Used by: EC_SALUD_OCUPACIONAL

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,6})	EC_CM_ALFANUM
otherwise	EC_APTITUD_LABORAL_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

EC_APTITUD_LABORAL_LISTA SECURE LOOKUP

Replaces the value with one of the 4 values in ec-aptitud-laboral.txt , chosen by a keyed hash.

Used by: EC_APTITUD_LABORAL

CONFIGURATION

lookupFile	ec-aptitud-laboral.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_AUTOIDENTIFICACION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_AUTOIDENTIFICACION_LISTA .

Used by: EC_L3_ETNIA

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,6})	EC_AUTOIDENTIFICACION_CODIGO
otherwise	EC_AUTOIDENTIFICACION_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

EC_AUTOIDENTIFICACION_CODIGO SECURE LOOKUP

Replaces the value with one of the 8 values in ec-autoidentificacion-codigos.txt , chosen by a keyed hash.

Used by: EC_AUTOIDENTIFICACION

CONFIGURATION

lookupFile	ec-autoidentificacion-codigos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_AUTOIDENTIFICACION_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in ec-autoidentificacion.txt , chosen by a keyed hash.

Used by: EC_AUTOIDENTIFICACION

CONFIGURATION

lookupFile	ec-autoidentificacion.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: EC_CATEGORIA_SUPRIMIDA , EC_ESTADO_CIVIL , EC_NIVEL_INSTRUCCION , EC_AUTOIDENTIFICACION , EC_IDENTIDAD_CULTURAL , EC_LENGUA , EC_CONDICION_MIGRATORIA , EC_RELIGION , EC_IDEOLOGIA , EC_PARTIDO , EC_ORIENTACION_SEXUAL , EC_IDENTIDAD_GENERO , EC_COBERTURA_SALUD , EC_DIAGNOSTICO , EC_MEDICAMENTO , EC_DISCAPACIDAD , EC_APTITUD_LABORAL , EC_HISTORIAL_CREDITICIO , EC_PROGRAMA_SOCIAL , EC_REGISTRO_SOCIAL , EC_TENENCIA_VIVIENDA , EC_ANTECEDENTES

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no)	EC_BANDERA_SI_NO
(?i)(true false)	EC_BANDERA_TRUE_FALSE
(?i)(yes)	EC_BANDERA_YES_NO

(?i)([sn])	EC_BANDERA_S_N
(?i)([yx])	EC_BANDERA_Y_N
([01])	EC_BANDERA_0_1
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.

Used by: EC_BANDERA

CONFIGURATION

lookupFile	ec-bandera-0-1.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.

Used by: EC_BANDERA

CONFIGURATION

lookupFile	ec-bandera-s-n.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt` , chosen by a keyed hash.

Used by: EC_BANDERA

CONFIGURATION

lookupFile	ec-bandera-si-no.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt` , chosen by a keyed hash.

Used by: EC_BANDERA

CONFIGURATION

lookupFile	ec-bandera-true-false.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-y-n.txt , chosen by a keyed hash.

Used by: EC_BANDERA

CONFIGURATION

lookupFile	ec-bandera-y-n.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in ec-bandera-yes-no.txt , chosen by a keyed hash.

Used by: EC_BANDERA

CONFIGURATION

lookupFile	ec-bandera-yes-no.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_CATEGORIA_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_SUPRIMIR .

Used by: EC_L3_AFILIACION_SINDICAL , EC_L3_VICTIMA , EC_SALUD_VIH , EC_SALUD_REPRODUCTIVA , EC_SALUD_MENTAL

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,6})	EC_CM_ALFANUM
otherwise	EC_SUPRIMIR

CONFIGURATION

trimInput	yes
requireMask	no

EC_CEDULA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_CM_DIGITOS .

Used by: EC_DOCUMENTO

PATTERNS, TRIED IN ORDER

((?:[0-2]\d 30)[0-5]\d{7})	EC_CEDULA_LUHN
((?:[0-2]\d 30)[0-5]\d{6})([\s-])(\d)	EC_CM_DIGITOS · keep · EC_CM_DIGITOS
otherwise	EC_CM_DIGITOS

CONFIGURATION

trimInput	yes
requireMask	no

EC_CEDULA_LUHN PAYMENT CARD

Replaces the card number with another that passes the Luhn check.

Used by: EC_CEDULA , EC_RUC

CONFIGURATION

minMaskedPositions	1
preserve	3

EC_CIE10 SECURE LOOKUP

Replaces the value with one of the 37 values in ec-cie10.txt , chosen by a keyed hash.

Used by: EC_DIAGNOSTICO

CONFIGURATION

lookupFile	ec-cie10.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_CIE10_DECIMAL SECURE LOOKUP

Replaces the value with one of the 33 values in ec-cie10-decimal.txt , chosen by a keyed hash.

Used by: EC_DIAGNOSTICO

CONFIGURATION

lookupFile	ec-cie10-decimal.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_CIUDADELA SECURE LOOKUP

Replaces the value with one of the 108 values in ec-ciudadeslas.txt , chosen by a keyed hash.

Used by: EC_L2_CIUDADELA

CONFIGURATION

lookupFile	ec-ciudadeslas.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE

inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

Used by: EC_L1_DOCUMENTO_EXTRANJERO , EC_L1_PASAPORTE , EC_L1_ACTA , EC_L1_SEGURO_SOCIAL , EC_L1_CODIGO_ESTUDIANTE , EC_L1_REGISTRO_PROFESIONAL , EC_L1_DIRECCION_COMPLEMENTO , EC_L1_BILLETERA , EC_L1_COOKIE , EC_L1_VEHICULO , EC_L1_INMUEBLE , EC_L1_CONTRATO , EC_L1_USUARIO , EC_SALUD_IDENTIFICADOR , EC_SALUD_RESULTADO_EXAMEN , EC_CATEGORIA_SUPRIMIDA , EC_DOCUMENTO , EC_NOMBRES , EC_APELLIDOS , EC_NOMBRE_COMPLETO , EC_EMAIL , EC_IDENTIDAD_CULTURAL , EC LENGUA , EC_CONDICION_MIGRATORIA , EC_RELIGION , EC_IDEOLOGIA , EC_PARTIDO , EC_ORIENTACION_SEXUAL , EC_IDENTIDAD_GENERO , EC_MEDICAMENTO , EC_DISCAPACIDAD , EC_APTITUD_LABORAL , EC_PROGRAMA_SOCIAL , EC_TENENCIA_VIVIENDA , EC_ANTECEDENTES , EC_EXPEDIENTE_JUDICIAL

CONFIGURATION

characterGroups	0123456789 AEIOU BCFGHJKLMNPQRSTUVWXYZ aeiou bcdfghjklmnpqrstvwxyz ÁÉÍÓÚÚ áéíóúú
caseSensitive	yes
minMaskedPositions	0

EC_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

Used by: EC_L1_CUENTA_BANCARIA , EC_CEDULA , EC_RUC , EC_TELEFONO , EC_COORDENADA , EC_ANIO , EC_EDAD , EC_OCUPACION_O_CODIGO , EC_COBERTURA_SALUD , EC_PROCEDIMIENTO , EC_HISTORIAL_CREDITICIO , EC_EXPEDIENTE_JUDICIAL

CONFIGURATION

characterGroups	0123456789
caseSensitive	yes
minMaskedPositions	0

EC_CM_DIGITOS_LETRAS CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

Used by: EC_L1_PLACA

CONFIGURATION

characterGroups	0123456789 ABCDEFGHIJKLMNOPQRSTUVWXYZ
caseSensitive	yes
minMaskedPositions	0

EC_CM_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

Used by: EC_L1_DISPOSITIVO , EC_IP

CONFIGURATION

characterGroups	0123456789 abcdef ABCDEF ghijklmnopqrstuvwxyz GHIJKLMNOPQRSTUVWXYZ
caseSensitive	yes
minMaskedPositions	0

EC_COBERTURA SECURE LOOKUP

Replaces the value with one of the 8 values in ec-cobertura-salud.txt , chosen by a keyed hash.

Used by: EC_COBERTURA_SALUD

CONFIGURATION

lookupFile	ec-cobertura-salud.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_COBERTURA_SALUD REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_COBERTURA .

Used by: EC_SALUD_COBERTURA

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,8})	EC_CM_DIGITOS
otherwise	EC_COBERTURA

CONFIGURATION

trimInput	yes
requireMask	no

EC_CODIGO_POSTAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: EC_L2_CODIGO_POSTAL

PATTERNS, TRIED IN ORDER

(\d{2})(\d{4})	keep · replace with 0000
(\d{2})(\d{3})	keep · replace with 000
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

EC_CONDICION_MIGRATORIA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_CONDICION_MIGRATORIA_LISTA .

Used by: EC_L3_CONDICION_MIGRATORIA

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,6})	EC_CM_ALFANUM
otherwise	EC_CONDICION_MIGRATORIA_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

EC_CONDICION_MIGRATORIA_LISTA SECURE LOOKUP

Replaces the value with one of the 10 values in `ec-condicion-migratoria.txt` , chosen by a keyed hash.

Used by: `EC_CONDICION_MIGRATORIA`

CONFIGURATION

lookupFile	ec-condicion-migratoria.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_CONTEO_LIMITADO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `EC_L2_PERSONAS_A_CARGO`

PATTERNS, TRIED IN ORDER

<code>([0-5])</code>	keep
<code>(\d+)</code>	replace with 5
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

EC_COORDENADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `EC_L1_GEOLOCALIZACION`

PATTERNS, TRIED IN ORDER

<code>(-?\d{1,3}[.,]\d)(\d+)\s*([,;])\s*(-?\d{1,3}[.,]\d)(\d+)</code>	keep · <code>EC_CM_DIGITOS</code> · keep · keep · <code>EC_CM_DIGITOS</code>
<code>(-?\d{1,3}[.,]\d)(\d+)</code>	keep · <code>EC_CM_DIGITOS</code>
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

EC_DIAGNOSTICO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches:

`EC_DIAGNOSTICO_TEXTO` .

Used by: `EC_SALUD_DIAGNOSTICO`

PATTERNS, TRIED IN ORDER

<code>([A-TV-Za-tv-z]\d{2}\.\d{1,2})</code>	<code>EC_CIE10_DECIMAL</code>
<code>([A-TV-Za-tv-z]\d{2,3}X?)</code>	<code>EC_CIE10</code>
<code>(?i)(s[ií] no true false yes [sn]yx01)</code>	<code>EC_BANDERA</code>
otherwise	<code>EC_DIAGNOSTICO_TEXTO</code>

CONFIGURATION

trimInput	yes
requireMask	no

EC_DIAGNOSTICO_TEXTO SECURE LOOKUP

Replaces the value with one of the 30 values in `ec-diagnostics.txt` , chosen by a keyed hash.

Used by: `EC_DIAGNOSTICO`

CONFIGURATION

lookupFile	<code>ec-diagnostics.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_DIRECCION SECURE LOOKUP

Replaces the value with one of the 4,000 values in `ec-direcciones.txt` , chosen by a keyed hash.

Used by: `EC_L1_DIRECCION`

CONFIGURATION

lookupFile	<code>ec-direcciones.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_DISCAPACIDAD REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`EC_DISCAPACIDAD_LISTA` .

Used by: `EC_SALUD_DISCAPACIDAD`

PATTERNS, TRIED IN ORDER

<code>(?i)(s[ií] no true false yes [snyx01])</code>	<code>EC_BANDERA</code>
<code>(\d{1,6})</code>	<code>EC_CM_ALFANUM</code>
otherwise	<code>EC_DISCAPACIDAD_LISTA</code>

CONFIGURATION

trimInput	yes
requireMask	no

EC_DISCAPACIDAD_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `ec-discapacidades.txt` , chosen by a keyed hash.

Used by: `EC_DISCAPACIDAD`

CONFIGURATION

lookupFile	<code>ec-discapacidades.txt</code>
hashMethod	SHA256

maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_DOCUMENTO REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: EC_CM_ALFANUM .

Used by: EC_L1_CEDULA

PATTERNS, TRIED IN ORDER

((?:[0-2]\d 30)[0-5]\d{7}00\d)	EC_RUC
(\d{2}[69]\d{10})	keep
((?:[0-2]\d 30)[0-5]\d{7})	EC_CEDULA
((?:[0-2]\d 30)[0-5]\d{6}[\s-]\d)	EC_CEDULA
otherwise	EC_CM_ALFANUM

CONFIGURATION

trimInput	yes
requireMask	no

EC_DPA DATA CLEANSING

Replaces each value found in ec-dpa-generalizado.txt (1,006 pairs) by its counterpart; values not in the file stay as they are.

Used by: EC_L2_DPA

CONFIGURATION

lookupFile	ec-dpa-generalizado.txt
delimiter	
caseSensitive	yes
trimWhitespace	yes

EC_EDAD REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: EC_L2_EDAD

PATTERNS, TRIED IN ORDER

(\d{3,})	replace with 90
([1-9])(\d)([.,]\d+)	keep · EC_CM_DIGITOS · keep
([1-9])(\d)	keep · EC_CM_DIGITOS
(\d)	EC_CM_DIGITOS
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

EC_EMAIL REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: EC_CM_ALFANUM .

Used by: EC_L1_EMAIL

PATTERNS, TRIED IN ORDER

(([^@\\s]+)@([^[^\\s]+)\\.([A-Za-z]{2,24})(?:\\.([A-Za-z]{2})?)) EC_CM_ALFANUM · EC_CM_ALFANUM · replace with test

otherwise EC_CM_ALFANUM

CONFIGURATION

trimInput yes

requireMask no

EC_EMPLEADOR SECURE LOOKUP

Replaces the value with one of the 24 values in ec-empleadores.txt , chosen by a keyed hash.

Used by: EC_L2_EMPLEADOR

CONFIGURATION

lookupFile ec-empleadores.txt

hashMethod SHA256

maskedValueCase PRESERVE_LOOKUP_FILE

inputCaseSensitive no

trimWhitespaceFromInput yes

trimWhitespaceInLookupFile yes

EC_ESTADO_CIVIL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_ESTADO_CIVIL_LISTA ·

Used by: EC_L2_ESTADO_CIVIL

PATTERNS, TRIED IN ORDER

(?i)(s[ií]|no|true|false|yes|[snyx01]) EC_BANDERA

(\\d{1,6}) EC_ESTADO_CIVIL_CODIGO

otherwise EC_ESTADO_CIVIL_LISTA

CONFIGURATION

trimInput yes

requireMask no

EC_ESTADO_CIVIL_CODIGO SECURE LOOKUP

Replaces the value with one of the 6 values in ec-estado-civil-codigos.txt , chosen by a keyed hash.

Used by: EC_ESTADO_CIVIL

CONFIGURATION

lookupFile ec-estado-civil-codigos.txt

hashMethod SHA256

maskedValueCase PRESERVE_INPUT

inputCaseSensitive no

trimWhitespaceFromInput yes

trimWhitespaceInLookupFile yes

EC_ESTADO_CIVIL_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in ec-estado-civil.txt , chosen by a keyed hash.

Used by: EC_ESTADO_CIVIL

CONFIGURATION

lookupFile	ec-estado-civil.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_ESTADO_CREDITO SECURE LOOKUP

Replaces the value with one of the 15 values in ec-estados-credito.txt , chosen by a keyed hash.

Used by: EC_HISTORIAL_CREDITICIO

CONFIGURATION

lookupFile	ec-estados-credito.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_EXPEDIENTE_JUDICIAL REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: EC_CM_ALFANUM .

Used by: EC_PENAL_EXPEDIENTE

PATTERNS, TRIED IN ORDER

(\d{5})(-)(\d{4})(-)(\d{5})	EC_CM_DIGITOS · keep · keep · keep · EC_CM_DIGITOS
(\d{1,6})([\s\-\-])([A-Za-z]{2,10})([\s\-\-])(\d{4})	EC_CM_DIGITOS · keep · keep · keep · keep
([A-Za-z]{1,6}[\s\-\-?])(\d{1,6})([\s\-\-])(\d{4})	keep · EC_CM_DIGITOS · keep · keep
(\d{1,6})([\s\-\-])(\d{4})	EC_CM_DIGITOS · keep · keep
otherwise	EC_CM_ALFANUM

CONFIGURATION

trimInput	yes
requireMask	no

EC_FECHA_EVENTO DATE SHIFT

Moves the date by -30 to 30 days.

Used by: EC_L2_FECHA_EVENTO

CONFIGURATION

minRange	-30
maxRange	30
unit	DAYS
roll	no

EC_FECHA_NACIMIENTO DATE SHIFT

Moves the date by -60 to 60 days.

Used by: EC_L2_FECHA_NACIMIENTO

CONFIGURATION

minRange	-60
maxRange	60
unit	DAYS
roll	no

EC_GRUPO_SANGUINEO SECURE LOOKUP

Replaces the value with one of the 8 values in ec-grupos-sanguineos.txt , chosen by a keyed hash.

Used by: EC_SALUD_GRUPO_SANGUINEO

CONFIGURATION

lookupFile	ec-grupos-sanguineos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_HISTORIAL_CREDITICIO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_ESTADO_CREDITO .

Used by: EC_FIN_HISTORIAL_CREDITICIO

PATTERNS, TRIED IN ORDER

(\d{1,4})	EC_CM_DIGITOS
(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
otherwise	EC_ESTADO_CREDITO

CONFIGURATION

trimInput	yes
requireMask	no

EC_IDENTIDAD_CULTURAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_IDENTIDAD_CULTURAL_LISTA .

Used by: EC_L3_IDENTIDAD_CULTURAL

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,6})	EC_CM_ALFANUM
otherwise	EC_IDENTIDAD_CULTURAL_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

EC_IDENTIDAD_CULTURAL_LISTA SECURE LOOKUP

Replaces the value with one of the 30 values in `ec-identidad-cultural.txt` , chosen by a keyed hash.

Used by: `EC_IDENTIDAD_CULTURAL`

CONFIGURATION

lookupFile	ec-identidad-cultural.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_IDENTIDAD_GENERO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`EC_IDENTIDAD_GENERO_LISTA` .

Used by: `EC_L3_IDENTIDAD_GENERO`

PATTERNS, TRIED IN ORDER

<code>(?i)(s[ií] no true false yes [snyx01])</code>	<code>EC_BANDERA</code>
<code>(\d{1,6})</code>	<code>EC_CM_ALFANUM</code>
otherwise	<code>EC_IDENTIDAD_GENERO_LISTA</code>

CONFIGURATION

trimInput	yes
requireMask	no

EC_IDENTIDAD_GENERO_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `ec-identidades-genero.txt` , chosen by a keyed hash.

Used by: `EC_IDENTIDAD_GENERO`

CONFIGURATION

lookupFile	ec-identidades-genero.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_IDEOLOGIA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`EC_IDEOLOGIA_LISTA` .

Used by: `EC_L3_IDEOLOGIA`

PATTERNS, TRIED IN ORDER

<code>(?i)(s[ií] no true false yes [snyx01])</code>	<code>EC_BANDERA</code>
<code>(\d{1,6})</code>	<code>EC_CM_ALFANUM</code>
otherwise	<code>EC_IDEOLOGIA_LISTA</code>

CONFIGURATION

trimInput	yes
requireMask	no

EC_IDEOLOGIA_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in `ec-ideologias.txt` , chosen by a keyed hash.

Used by: `EC_IDEOLOGIA`

CONFIGURATION

lookupFile	<code>ec-ideologias.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_INGRESOS NUMERIC MAPPING

Maps the number to another between 450 and 9,000.

Used by: `EC_FIN_INGRESOS`

CONFIGURATION

minValue	<code>450</code>
maxValue	<code>9000</code>

EC_INSTITUCION_EDUCATIVA SECURE LOOKUP

Replaces the value with one of the 12 values in `ec-instituciones-educativas.txt` , chosen by a keyed hash.

Used by: `EC_L2_INSTITUCION_EDUCATIVA`

CONFIGURATION

lookupFile	<code>ec-instituciones-educativas.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_IP REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `EC_CM_HEX` .

Used by: `EC_L1_IP`

PATTERNS, TRIED IN ORDER

<code>(\d{1,3})\.(\d{1,3})\.(\d{1,3})\.(\d{1,3})</code>	<code>EC_OCTETO</code> · <code>EC_OCTETO</code> · <code>EC_OCTETO</code> · <code>EC_OCTETO</code>
otherwise	<code>EC_CM_HEX</code>

CONFIGURATION

trimInput	yes
requireMask	no

EC_LENGUA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_LENGUA_LISTA .

Used by: EC_L3_LENGUA

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,6})	EC_CM_ALFANUM
otherwise	EC_LENGUA_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

EC_LENGUA_LISTA SECURE LOOKUP

Replaces the value with one of the 16 values in ec-lenguas.txt , chosen by a keyed hash.

Used by: EC_LENGUA

CONFIGURATION

lookupFile	ec-lenguas.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_MEDICAMENTO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_MEDICAMENTO_LISTA .

Used by: EC_SALUD_MEDICAMENTO

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,6})	EC_CM_ALFANUM
otherwise	EC_MEDICAMENTO_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

EC_MEDICAMENTO_LISTA SECURE LOOKUP

Replaces the value with one of the 22 values in ec-medicamentos.txt , chosen by a keyed hash.

Used by: EC_MEDICAMENTO

CONFIGURATION

lookupFile	ec-medicamentos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes

trimWhitespaceInLookupFile	yes
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EC_NACIONALIDAD_PAIS SECURE LOOKUP

Replaces the value with one of the 14 values in `ec-nacionalidades.txt` , chosen by a keyed hash.

Used by: `EC_L3_NACIONALIDAD`

CONFIGURATION

lookupFile	<code>ec-nacionalidades.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	<code>no</code>
trimWhitespaceFromInput	<code>yes</code>
trimWhitespaceInLookupFile	<code>yes</code>

EC_NIVEL_INSTRUCCION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`EC_NIVEL_INSTRUCCION_LISTA` .

Used by: `EC_L2_NIVEL_INSTRUCCION`

PATTERNS, TRIED IN ORDER

<code>(?i)(s[ií] no true false yes [snyx01])</code>	<code>EC_BANDERA</code>
<code>(\d{1,6})</code>	<code>EC_NIVEL_INSTRUCCION_CODIGO</code>
otherwise	<code>EC_NIVEL_INSTRUCCION_LISTA</code>

CONFIGURATION

trimInput	<code>yes</code>
requireMask	<code>no</code>

EC_NIVEL_INSTRUCCION_CODIGO SECURE LOOKUP

Replaces the value with one of the 8 values in `ec-nivel-instruccion-codigos.txt` , chosen by a keyed hash.

Used by: `EC_NIVEL_INSTRUCCION`

CONFIGURATION

lookupFile	<code>ec-nivel-instruccion-codigos.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	<code>no</code>
trimWhitespaceFromInput	<code>yes</code>
trimWhitespaceInLookupFile	<code>yes</code>

EC_NIVEL_INSTRUCCION_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `ec-nivel-instruccion.txt` , chosen by a keyed hash.

Used by: `EC_NIVEL_INSTRUCCION`

CONFIGURATION

lookupFile	<code>ec-nivel-instruccion.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	<code>no</code>

trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_NOMBRE NAME

Replaces the name with one of the 338 names in `ec-nombres.txt` .

Used by: `EC_PALABRA_NOMBRE`

CONFIGURATION

lookupFile	<code>ec-nombres.txt</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
filterAccent	yes
inputCaseSensitive	no

EC_NOMBRE_COMPLETO REGEX DECOMPOSE

Tries 7 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_ALFANUM` .

Used by: `EC_L1_NOMBRE_COMPLETO`

PATTERNS, TRIED IN ORDER

<code>([^\,]+),\s*(.+)</code>	<code>EC_APELLIDOS</code> · <code>EC_NOMBRES</code>
<code>(\S+)</code>	<code>EC_PALABRA_NOMBRE</code>
<code>(\S+)\s+(\S+)</code>	<code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_APELLIDO</code>
<code>(\S+)\s+(\S+)\s+(\S+)</code>	<code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_APELLIDO</code> · <code>EC_PALABRA_APELLIDO</code>
<code>(\S+)\s+(\S+)\s+(\S+)\s+(\S+)</code>	<code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_APELLIDO</code> · <code>EC_PALABRA_APELLIDO</code>
<code>(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)</code>	<code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_APELLIDO</code> · <code>EC_PALABRA_APELLIDO</code> · <code>EC_PALABRA_APELLIDO</code>
<code>(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)</code>	<code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_APELLIDO</code> · <code>EC_PALABRA_APELLIDO</code> · <code>EC_PALABRA_APELLIDO</code>
otherwise	<code>EC_CM_ALFANUM</code>

CONFIGURATION

trimInput	yes
requireMask	no

EC_NOMBRES REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_ALFANUM` .

Used by: `EC_L1_NOMBRE` , `EC_NOMBRE_COMPLETO`

PATTERNS, TRIED IN ORDER

<code>(\S+)</code>	<code>EC_PALABRA_NOMBRE</code>
<code>(\S+)\s+(\S+)</code>	<code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_NOMBRE</code>
<code>(\S+)\s+(\S+)\s+(\S+)</code>	<code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_NOMBRE</code>
<code>(\S+)\s+(\S+)\s+(\S+)\s+(\S+)</code>	<code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_NOMBRE</code> · <code>EC_PALABRA_NOMBRE</code>
otherwise	<code>EC_CM_ALFANUM</code>

CONFIGURATION

trimInput	yes
requireMask	no

EC_OCTETO NUMERIC MAPPING

Maps the number to another between 1 and 254.

Used by: EC_IP

CONFIGURATION

minValue	1
maxValue	254

EC_OCUPACION SECURE LOOKUP

Replaces the value with one of the 38 values in ec-ocupaciones.txt , chosen by a keyed hash.

Used by: EC_OCUPACION_0_CODIGO

CONFIGURATION

lookupFile	ec-ocupaciones.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_OCUPACION_0_CODIGO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: EC_OCUPACION .

Used by: EC_L2_OCUPACION

PATTERNS, TRIED IN ORDER

(\d{2,5})	EC_CM_DIGITOS
otherwise	EC_OCUPACION

CONFIGURATION

trimInput	yes
requireMask	no

EC_ORIENTACION_SEXUAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_ORIENTACION_SEXUAL_LISTA .

Used by: EC_L3_ORIENTACION_SEXUAL

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,6})	EC_CM_ALFANUM
otherwise	EC_ORIENTACION_SEXUAL_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

EC_ORIENTACION_SEXUAL_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in ec-orientaciones-sexuales.txt , chosen by a keyed hash.

Used by: EC_ORIENTACION_SEXUAL

CONFIGURATION

lookupFile	ec-orientaciones-sexuales.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_PALABRA_APELLIDO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: EC_APELLIDO .

Used by: EC_APELLIDOS , EC_NOMBRE_COMPLETO

PATTERNS, TRIED IN ORDER

(?i)	keep
(de del la las los y e san santa van von di da)	
otherwise	EC_APELLIDO

CONFIGURATION

trimInput	yes
requireMask	no

EC_PALABRA_NOMBRE REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: EC_NOMBRE .

Used by: EC_NOMBRES , EC_NOMBRE_COMPLETO

PATTERNS, TRIED IN ORDER

(?i)	keep
(de del la las los y e san santa van von di da)	
otherwise	EC_NOMBRE

CONFIGURATION

trimInput	yes
requireMask	no

EC_PARROQUIA DATA CLEANSING

Replaces each value found in ec-parroquias-generalizadas.txt (25,214 pairs) by its counterpart; values not in the file stay as they are.

Used by: EC_L2_PARROQUIA

CONFIGURATION

lookupFile	ec-parroquias-generalizadas.txt
delimiter	
caseSensitive	yes
trimWhitespace	yes

EC_PARTIDO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_PARTIDO_LISTA .

Used by: EC_L3_FILIACION_POLITICA

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,6})	EC_CM_ALFANUM
otherwise	EC_PARTIDO_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

EC_PARTIDO_LISTA SECURE LOOKUP

Replaces the value with one of the 13 values in `ec-partidos.txt` , chosen by a keyed hash.

Used by: `EC_PARTIDO`

CONFIGURATION

lookupFile	ec-partidos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_PATRIMONIO NUMERIC MAPPING

Maps the number to another between 1,000 and 600,000.

Used by: `EC_FIN_PATRIMONIO`

CONFIGURATION

minValue	1000
maxValue	600000

EC_PROCEDIMIENTO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches:

`EC_PROCEDIMIENTO_TEXTO` .

Used by: `EC_SALUD_PROCEDIMIENTO`

PATTERNS, TRIED IN ORDER

(\d{4,6})	EC_CM_DIGITOS
otherwise	EC_PROCEDIMIENTO_TEXTO

CONFIGURATION

trimInput	yes
requireMask	no

EC_PROCEDIMIENTO_TEXTO SECURE LOOKUP

Replaces the value with one of the 10 values in `ec-procedimientos.txt` , chosen by a keyed hash.

Used by: `EC_PROCEDIMIENTO`

CONFIGURATION

lookupFile	ec-procedimientos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT

inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_PROGRAMA_SOCIAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_PROGRAMA_SOCIAL_LISTA .

Used by: EC_FIN_PROGRAMA_SOCIAL

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,6})	EC_CM_ALFANUM
otherwise	EC_PROGRAMA_SOCIAL_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

EC_PROGRAMA_SOCIAL_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in ec-programas-sociales.txt , chosen by a keyed hash.

Used by: EC_PROGRAMA_SOCIAL

CONFIGURATION

lookupFile	ec-programas-sociales.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_RAZON_SOCIAL SECURE LOOKUP

Replaces the value with one of the 20 values in ec-razones-sociales.txt , chosen by a keyed hash.

Used by: EC_L1_RAZON_SOCIAL

CONFIGURATION

lookupFile	ec-razones-sociales.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_REDACTAR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: EC_L1_CREDENCIAL , EC_L3_BIOMETRICO , EC_L3_GENETICO

PATTERNS, TRIED IN ORDER

(.+)	each character → X
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

EC_REGISTRO_SOCIAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_REGISTRO_SOCIAL_LISTA .

Used by: EC_FIN_REGISTRO_SOCIAL

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,6})	EC_REGISTRO_SOCIAL_CODIGO
otherwise	EC_REGISTRO_SOCIAL_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

EC_REGISTRO_SOCIAL_CODIGO SECURE LOOKUP

Replaces the value with one of the 5 values in ec-registro-social-codigos.txt , chosen by a keyed hash.

Used by: EC_REGISTRO_SOCIAL

CONFIGURATION

lookupFile	ec-registro-social-codigos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_REGISTRO_SOCIAL_LISTA SECURE LOOKUP

Replaces the value with one of the 9 values in ec-registro-social.txt , chosen by a keyed hash.

Used by: EC_REGISTRO_SOCIAL

CONFIGURATION

lookupFile	ec-registro-social.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_RELIGION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_RELIGION_LISTA .

Used by: EC_L3_RELIGION

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,6})	EC_CM_ALFANUM

otherwise	EC_RELIGION_LISTA
CONFIGURATION	
trimInput	yes
requireMask	no

EC_RELIGION_LISTA SECURE LOOKUP

Replaces the value with one of the 10 values in `ec-religiones.txt` , chosen by a keyed hash.

Used by: `EC_RELIGION`

CONFIGURATION	
lookupFile	<code>ec-religiones.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_RUC REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_DIGITOS` .

Used by: `EC_L1_RUC` , `EC_DOCUMENTO`

PATTERNS, TRIED IN ORDER	
<code>((?:[0-2]\d 30)[0-5]\d{7})(00\d)</code>	<code>EC_CEDULA_LUHN</code> · keep
<code>(\d{2}[69]\d{10})</code>	keep
otherwise	<code>EC_CM_DIGITOS</code>

CONFIGURATION	
trimInput	yes
requireMask	no

EC_SEXO REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `EC_L2_SEXO`

PATTERNS, TRIED IN ORDER	
<code>(?i)(femenino masculino)</code>	<code>EC_SEXO_FEMENINO_MASCULINO</code>
<code>(?i)(mujer hombre)</code>	<code>EC_SEXO_MUJER_HOMBRE</code>
<code>(?i)(female male)</code>	<code>EC_SEXO_FEMALE_MALE</code>
<code>(?i)([fm])</code>	<code>EC_SEXO_F_M</code>
<code>(?i)([hm])</code>	<code>EC_SEXO_H_M</code>
<code>([12])</code>	<code>EC_SEXO_1_2</code>
otherwise	keep

CONFIGURATION	
trimInput	yes
requireMask	no

EC_SEX0_1_2 SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-sexo-1-2.txt` , chosen by a keyed hash.

Used by: `EC_SEX0`

CONFIGURATION

lookupFile	ec-sexo-1-2.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_SEX0_F_M SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-sexo-f-m.txt` , chosen by a keyed hash.

Used by: `EC_SEX0`

CONFIGURATION

lookupFile	ec-sexo-f-m.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_SEX0_FEMALE_MALE SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-sexo-female-male.txt` , chosen by a keyed hash.

Used by: `EC_SEX0`

CONFIGURATION

lookupFile	ec-sexo-female-male.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_SEX0_FEMENINO_MASCULINO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-sexo-femenino-masculino.txt` , chosen by a keyed hash.

Used by: `EC_SEX0`

CONFIGURATION

lookupFile	ec-sexo-femenino-masculino.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_SEXO_H_M SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-sexo-h-m.txt` , chosen by a keyed hash.

Used by: `EC_SEXO`

CONFIGURATION

lookupFile	ec-sexo-h-m.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_SEXO_MUJER_HOMBRE SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-sexo-mujer-hombre.txt` , chosen by a keyed hash.

Used by: `EC_SEXO`

CONFIGURATION

lookupFile	ec-sexo-mujer-hombre.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in `ec-sin-informacion.txt` , chosen by a keyed hash.

Used by: `EC_CATEGORIA_SUPRIMIDA`

CONFIGURATION

lookupFile	ec-sin-informacion.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_TARJETA REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `EC_L1_TARJETA`

PATTERNS, TRIED IN ORDER

<code>(\d[\d\s\~]{11,22}\d)</code>	<code>EC_TARJETA_LUHN</code>
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

EC_TARJETA_LUHN PAYMENT CARD

Replaces the card number with another that passes the Luhn check.

Used by: EC_TARJETA

CONFIGURATION

minMaskedPositions	6
preserve	0

EC_TELEFONO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: EC_CM_DIGITOS .

Used by: EC_L1_TELEFONO

PATTERNS, TRIED IN ORDER

(\+?593[\s-]?) (0?9) ([\s-]?) ([\d\s-]{8,12})	keep · keep · keep · EC_CM_DIGITOS
(0?9) ([\s-]?) ([\d\s-]{8,12})	keep · keep · EC_CM_DIGITOS
(\+?593[\s-]?)? (\(0?[2-7]\)?) ([\s-]?) ([\d\s-]{6,10})	keep · keep · keep · EC_CM_DIGITOS
otherwise	EC_CM_DIGITOS

CONFIGURATION

trimInput	yes
requireMask	no

EC_TENENCIA_VIVIENDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

EC_TENENCIA_VIVIENDA_LISTA .

Used by: EC_FIN_VIVIENDA

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false yes [snyx01])	EC_BANDERA
(\d{1,6})	EC_CM_ALFANUM
otherwise	EC_TENENCIA_VIVIENDA_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

EC_TENENCIA_VIVIENDA_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in ec-tenencia-vivienda.txt , chosen by a keyed hash.

Used by: EC_TENENCIA_VIVIENDA

CONFIGURATION

lookupFile	ec-tenencia-vivienda.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

EC_TEXTO_LIBRE FREE TEXT REDACTION

Replaces the text matching 8 patterns with [DATO] , and the 1,668 terms of ec-texto-nombres.txt with [NOMBRE] .

Used by: EC_L1_TEXTO_LIBRE , EC_SALUD_TEXTO_CLINICO

PATTERNS, TRIED IN ORDER

(?<!\d)(0[1-9] 1\d 2[0-4] 30)[0-5]\d{7}00\d(?:\d)
(?<!\d)(0[1-9] 1\d 2[0-4] 30)[0-5]\d{7}(?:\d)
(?<!\d)\d{9}-\d(?:\d)
[\w.+\\-]+@[\\w\\-]+(\\. [\\w\\-]+)+
(?<![\\d\\-])(\\+?593[\\s\\-]?)?0?9\\d{2}[\\s\\-]?\\d{3}[\\s\\-]?\\d{3}(?![\\d\\-])
(?<!\d)\d{13,19}(?:\d)
(?<![A-Za-z])[A-Z]{3}[\\s\\-]?\\d{3,4}(?![A-Za-z\\d])
(?<!\d)(\\d{1,3}\\.){3}\\d{1,3}(?:\d)

CONFIGURATION

regExRedactValue	[DATO]
lookupFile	ec-texto-nombres.txt
lookupFileRedactValue	[NOMBRE]
isDenyList	yes

EC_VALOR_DEUDA NUMERIC MAPPING

Maps the number to another between 100 and 250,000.

Used by: EC_FIN_DEUDA

CONFIGURATION

minValue	100
maxValue	250000

EC_VALOR_PENSION NUMERIC MAPPING

Maps the number to another between 200 and 3,000.

Used by: EC_FIN_PENSION

CONFIGURATION

minValue	200
maxValue	3000

APPENDIX B

Files

The lookup and detection files the set reads. On a Masking Engine each one is uploaded with the object that reads it — the lookup file of a Secure Lookup or Data Cleansing algorithm, the value list of a List classifier — and referenced by that object's configuration.

FILE	LINES	USED BY	FIRST LINES
ec- antecedentes.txt	7	EC_ANTECEDENTES_LISTA	No registra antecedentes penales · Registra antecedentes · Procesado · Sentenciado
ec-apellidos.txt	366	EC_APELLIDO	Zambrano · Vera · García · Mero
ec-aptitud- laboral.txt	4	EC_APTITUD_LABORAL_LISTA	Apto · Apto con restricciones · No apto · Pendiente
ec- autoidentificacio n-codigos.txt	8	EC_AUTOIDENTIFICACION_CODIGO	1 · 2 · 3 · 4
ec- autoidentificacio n.txt	8	EC_AUTOIDENTIFICACION_LISTA	Mestizo · Montubio · Afroecuatoriano · Negro
ec-bandera-0- 1.txt	2	EC_BANDERA_0_1	0 · 1
ec-bandera-s- n.txt	2	EC_BANDERA_S_N	S · N
ec-bandera-si- no.txt	2	EC_BANDERA_SI_NO	Sí · No
ec-bandera-true- false.txt	2	EC_BANDERA_TRUE_FALSE	true · false
ec-bandera-y- n.txt	2	EC_BANDERA_Y_N	Y · N
ec-bandera-yes- no.txt	2	EC_BANDERA_YES_NO	Yes · No
ec-cie10- decimal.txt	33	EC_CIE10_DECIMAL	E11.9 · E78.5 · J45.9 · J06.9
ec-cie10.txt	37	EC_CIE10	I10 · E11 · E78 · J45
ec- ciudadelas.txt	108	EC_CIUDADELA	Cdla. Kennedy · Cdla. Urdesa Central · Cdla. Urdesa Norte · Cdla. Alborada
ec-cobertura- salud.txt	8	EC_COBERTURA	IESS (Seguro General) · Seguro Social Campesino · ISSFA · ISSPOL
ec-condicion- migratoria.txt	10	EC_CONDICION_MIGRATORIA_LISTA	Ecuatoriano · Residente permanente · Residente temporal · Visitante temporal
ec-detectar- apellidos.txt	454	EC_L1_APELLIDO – List EC_L1_NOMBRE_COMPLETO – List	Zambrano · Vera · García · Garcia
ec-detectar- aptitud- laboral.txt	5	EC_SALUD_OCUPACIONAL – List	apto · no apto · apto con restricciones · pendiente

FILE	LINES	USED BY	FIRST LINES
ec-detectar-cobertura.txt	19	EC_SALUD_COBERTURA – List	IESS (Seguro General) · Seguro Social Campesino · ISSFA · ISSPOL
ec-detectar-codigos-banderas.txt	11	EC_L3_ETNIA – List EC_FIN_PROGRAMA_SOCIAL – List	s · n · sí · si
ec-detectar-condicion-migratoria.txt	22	EC_L3_CONDICION_MIGRATORIA – List	Ecuatoriano · Residente permanente · Residente temporal · Visitante temporal
ec-detectar-discapacidades.txt	27	EC_SALUD_DISCAPACIDAD – List	Física · Fisica · Auditiva · Visual
ec-detectar-estado-civil.txt	22	EC_L2_ESTADO_CIVIL – List	Soltero/a · Casado/a · Unión de hecho · Union de hecho
ec-detectar-etnia.txt	20	EC_L3_ETNIA – List	Mestizo · Montubio · Afroecuatoriano · Negro
ec-detectar-grupos-sanguineos.txt	18	EC_SALUD_GRUPO_SANGUINEO – List	A+ · A- · B+ · B-
ec-detectar-identidades-genero.txt	21	EC_L3_IDENTIDAD_GENERO – List	Mujer cisgénero · Mujer cisgenero · Hombre cisgénero · Hombre cisgenero
ec-detectar-ideologias.txt	16	EC_L3_IDEOLOGIA – List	Izquierda · Centroizquierda · Centro · Centroderecha
ec-detectar-lenguas.txt	29	EC_L3 LENGUA – List	Castellano · Kichwa · Shuar chicham · Achuar chicham
ec-detectar-medicamentos.txt	50	EC_SALUD_MEDICAMENTO – List	Paracetamol · Ibuprofeno · Diclofenaco · Metformina
ec-detectar-nacionalidades-indigenas.txt	67	EC_L3_IDENTIDAD_CULTURAL – List	Kichwa · Shuar · Achuar · Awá
ec-detectar-nacionalidades.txt	32	EC_L3_NACIONALIDAD – List	Ecuatoriana · Venezolana · Colombiana · Peruana
ec-detectar-nivel-instruccion.txt	25	EC_L2_NIVEL_INSTRUCCION – List	Ninguno · Centro de alfabetización · Centro de alfabetizacion · Educación inicial
ec-detectar-nombres.txt	395	EC_L1_NOMBRE – List EC_L1_NOMBRE_COMPLETO – List	José · Jose · María · Maria
ec-detectar-orientaciones-sexuales.txt	14	EC_L3_ORIENTACION_SEXUAL – List	Heterosexual · Gay · Lesbiana · Bisexual
ec-detectar-parroquias.txt	1,160	EC_L2_PARROQUIA – List	Guaranda · Facundo Vela · Julio E. Moreno (Catanahuan Grande) · San Lorenzo
ec-detectar-partidos.txt	25	EC_L3_FILIACION_POLITICA – List	Revolución Ciudadana · Revolucion Ciudadana · Acción Democrática Nacional · Accion Democratica Nacional

FILE	LINES	USED BY	FIRST LINES
ec-detectar-programas-sociales.txt	19	EC_FIN_PROGRAMA_SOCIAL - List	Bono de Desarrollo Humano · Bono Joaquín Gallegos Lara · Bono Joaquín Gallegos Lara · Pensión para Adultos Mayores
ec-detectar-registro-social.txt	19	EC_FIN_REGISTRO_SOCIAL - List	Quintil 1 · Quintil 2 · Quintil 3 · Quintil 4
ec-detectar-religiones.txt	35	EC_L3_RELIGION - List	Católica · Catolica · Evangélica · Evangelica
ec-detectar-sexo.txt	9	EC_L2_SEX0 - List	f · m · h · femenino
ec-detectar-sindicatos.txt	12	EC_L3_AFILIACION_SINDICAL - List	FUT · CEOSL · CEDOCUT · CTE
ec-detectar-tenencia-vivienda.txt	15	EC_FIN_VIVIENDA - List	Propia y totalmente pagada · Propia y la está pagando · Propia y la está pagando · Propia (regalada, donada, heredada)
ec-diagnosticos.txt	30	EC_DIAGNOSTICO_TEXTO	Hipertensión arterial · Diabetes mellitus tipo 2 · Asma · Lumbalgia
ec-direcciones.txt	4,000	EC_DIRECCION	Malecón Occidental 2067 y Simón Bolívar · Av. Fray Vicente Solano N52-293 y Mariscal Sucre · Barrio La Mariscal, Mz. M Villa 60 · Calle 10 de Agosto N38-214 y Brasil
ec-discapacidades.txt	8	EC_DISCAPACIDAD_LISTA	Física · Auditiva · Visual · Intelectual
ec-dpa-generalizado.txt	1,006	EC_DPA	020151 020150 · 020153 020150 · 020155 020150 · 020156 020150
ec-empleadores.txt	24	EC_EMPLEADOR	Distribuidora Andina Cía. Ltda. · Constructora Pichincha S.A. · Transportes Mitad del Mundo S.A. · Supermercados del Litoral S.A.
ec-estado-civil-codigos.txt	6	EC_ESTADO_CIVIL_CODIGO	1 · 2 · 3 · 4
ec-estado-civil.txt	6	EC_ESTADO_CIVIL_LISTA	Soltero/a · Casado/a · Unión de hecho · Separado/a
ec-estados-credito.txt	15	EC_ESTADO_CREDITO	A-1 Riesgo normal · A-2 Riesgo normal · A-3 Riesgo normal · B-1 Riesgo potencial
ec-grupos-sanguineos.txt	8	EC_GRUPO_SANGUINEO	A+ · A- · B+ · B-
ec-identidad-cultural.txt	30	EC_IDENTIDAD_CULTURAL_LISTA	Kichwa · Shuar · Achuar · Awá
ec-identidades-genero.txt	8	EC_IDENTIDAD_GENERO_LISTA	Mujer cisgénero · Hombre cisgénero · Mujer trans · Hombre trans
ec-ideologias.txt	7	EC_IDEOLOGIA_LISTA	Izquierda · Centroizquierda · Centro · Centroderecha
ec-instituciones-educativas.txt	12	EC_INSTITUCION_EDUCATIVA	Unidad Educativa Nuevo Amanecer · Unidad Educativa Fiscal 10 de Agosto · Escuela de Educación Básica Simón Bolívar · Colegio Nacional Mejía
ec-lenguas.txt	16	EC LENGUA_LISTA	Castellano · Kichwa · Shuar chicham · Achuar chicham
ec-medicamentos.txt	22	EC_MEDICAMENTO_LISTA	Paracetamol · Ibuprofeno · Diclofenaco · Metformina

FILE	LINES	USED BY	FIRST LINES
ec-nacionalidades.txt	14	EC_NACIONALIDAD_PAIS	Ecuatoriana · Venezolana · Colombiana · Peruana
ec-nivel-instruccion-codigos.txt	8	EC_NIVEL_INSTRUCCION_CODIGO	1 · 2 · 3 · 4
ec-nivel-instruccion.txt	8	EC_NIVEL_INSTRUCCION_LISTA	Ninguno · Centro de alfabetización · Educación inicial · Educación general básica
ec-nombres.txt	338	EC_NOMBRE	José · María · Juan · Carlos
ec-ocupaciones.txt	38	EC_OCUPACION	Empleado administrativo · Vendedor · Cajero · Chofer
ec-orientaciones-sexuales.txt	8	EC_ORIENTACION_SEXUAL_LISTA	Heterosexual · Gay · Lesbiana · Bisexual
ec-parroquias-generalizadas.txt	25,214	EC_PARROQUIA	Facundo Vela Guaranda · Facundo Vela - Guaranda Guaranda - Guaranda · Facundo Vela, Guaranda Guaranda, Guaranda · Facundo Vela (Guaranda) Guaranda (Guaranda)
ec-partidos.txt	13	EC_PARTIDO_LISTA	Revolución Ciudadana · Acción Democrática Nacional · Movimiento CREO · Partido Social Cristiano
ec-procedimientos.txt	100	EC_PROCEDIMIENTO_TEXT	Consulta médica general · Consulta de emergencia · Biometría hemática · Glucosa en sangre
ec-programas-sociales.txt	7	EC_PROGRAMA_SOCIAL_LISTA	Bono de Desarrollo Humano · Bono Joaquín Gallegos Lara · Pensión para Adultos Mayores · Pensión para Personas con Discapacidad
ec-razones-sociales.txt	20	EC_RAZON_SOCIAL	Comercial Andina Cía. Ltda. · Distribuidora del Pacífico S.A. · Importadora Los Andes Cía. Ltda. · Transportes Mitad del Mundo S.A.
ec-registro-social-codigos.txt	5	EC_REGISTRO_SOCIAL_CODIGO	1 · 2 · 3 · 4
ec-registro-social.txt	9	EC_REGISTRO_SOCIAL_LISTA	Quintil 1 · Quintil 2 · Quintil 3 · Quintil 4
ec-religiones.txt	10	EC_RELIGION_LISTA	Católica · Evangélica · Testigo de Jehová · Iglesia de Jesucristo de los Santos de los Últimos Días
ec-sexo-1-2.txt	2	EC_SEXO_1_2	1 · 2
ec-sexo-f-m.txt	2	EC_SEXO_F_M	F · M
ec-sexo-female-male.txt	2	EC_SEXO_FEMALE_MALE	Female · Male
ec-sexo-femenino-masculino.txt	2	EC_SEXO_FEMENINO_MASCULINO	Femenino · Masculino
ec-sexo-h-m.txt	2	EC_SEXO_H_M	H · M
ec-sexo-mujer-hombre.txt	2	EC_SEXO_MUJER_HOMBRE	Mujer · Hombre
ec-sin-informacion.txt	1	EC_SUPRIMIR	Sin información
ec-tenencia-vivienda.txt	7	EC_TENENCIA_VIVIENDA_LISTA	Propia y totalmente pagada · Propia y la está pagando · Propia (regalada, donada, heredada) · Prestada o cedida

FILE	LINES	USED BY	FIRST LINES
ec-texto- nombres.txt	1,668	EC_TEXT0_LIBRE	José · JOSÉ · Jose · JOSE

Exclusion lists

Words that keep a column-name rule from matching when they follow it — so that `NOMBRE_PRODUCTO` is not taken for a person's name.

E1 producto prod item articulo empresa emp razon social comercial archivo calle avenida pasaje ciudadela urbanizacion barrio parroquia canton provincia region depto dpto pais banco sucursal agencia plan campana proyecto servicio tabla columna campo usuario host servidor dominio marca modelo categoria tipo escuela colegio institucion universidad curso materia documento doc unidad clinica hospital iess seguro area dependencia cargo sistema app aplicacion grupo lista reporte informe proceso evento tarea perfil rol clase objeto cuenta medicamento diagnostico estudio programa beneficio partido sindicato religion pueblo nacionalidad lengua idioma lugar sede local tienda bodega proveedor convenio contrato poliza aseguradora entidad organismo organizacion parametro variable metodo funcion indice job script cola recurso red dispositivo plantilla imagen foto pagina sitio modulo menu opcion accion permiso concepto impuesto moneda emisor factura comprobante pago forma zona ruta linea actividad rubro sector file table column field user server product company category type school document department system group report role account place store supplier contract device image page module action key label code project city country street bank cliente cli paciente pac usuario afiliado asegurado beneficiario benef titular empleado trabajador colaborador alumno estudiante deudor codeudor garante solicitante propietario arrendatario inquilino conductor responsable representante apoderado persona ciudadano contribuyente madre padre conyuge conviviente contacto referencia victima denunciante procesado testigo medico docente socio heredero donante tutor jubilado pensionista derechohabiente customer patient employee member holder mother father spouse contact person student driver owner completo apellidos? usuario corto largo

E2 producto prod item articulo empresa emp razon social comercial archivo calle avenida pasaje ciudadela urbanizacion barrio parroquia canton provincia region depto dpto pais banco sucursal agencia plan campana proyecto servicio tabla columna campo usuario host servidor dominio marca modelo categoria tipo escuela colegio institucion universidad curso materia documento doc unidad clinica hospital iess seguro area dependencia cargo sistema app aplicacion grupo lista reporte informe proceso evento tarea perfil rol clase objeto cuenta medicamento diagnostico estudio programa beneficio partido sindicato religion pueblo nacionalidad lengua idioma lugar sede local tienda bodega proveedor convenio contrato poliza aseguradora entidad organismo organizacion parametro variable metodo funcion indice job script cola recurso red dispositivo plantilla imagen foto pagina sitio modulo menu opcion accion permiso concepto impuesto moneda emisor factura comprobante pago forma zona ruta linea actividad rubro sector file table column field user server product company category type school document department system group report role account place store supplier contract device image page module action key label code project city country street bank cliente cli paciente pac usuario afiliado asegurado beneficiario benef titular empleado trabajador colaborador alumno estudiante deudor codeudor garante solicitante propietario arrendatario inquilino conductor responsable representante apoderado persona ciudadano contribuyente madre padre conyuge conviviente contacto referencia victima denunciante procesado testigo medico docente socio heredero donante tutor jubilado pensionista derechohabiente customer patient employee member holder mother father spouse contact person student driver owner

E3	producto prod item articulo empresa emp razon social comercial archivo calle avenida pasaje ciudadela urbanizacion barrio parroquia canton provincia region depto dpto pais banco sucursal agencia plan campana proyecto servicio tabla columna campo usuario host servidor dominio marca modelo categoria tipo escuela colegio institucion universidad curso materia documento doc unidad clinica hospital iess seguro area dependencia cargo sistema app aplicacion grupo lista reporte informe proceso evento tarea perfil rol clase objeto cuenta medicamento diagnostico estudio programa beneficio partido sindicato religion pueblo nacionalidad lengua idioma lugar sede local tienda bodega proveedor convenio contrato poliza aseguradora entidad organismo organizacion parametro variable metodo funcion indice job script cola recurso red dispositivo plantilla imagen foto pagina sitio modulo menu opcion accion permiso concepto impuesto moneda emisor factura comprobante pago forma zona ruta linea actividad rubro sector file table column field user server product company category type school document department system group report role account place store supplier contract device image page module action key label code project city country street bank cliente cli paciente pac usuario afiliado asegurado beneficiario benef titular empleado trabajador colaborador alumno estudiante deudor codeudor garante solicitante propietario arrendatario inquilino conductor responsable representante apoderado persona ciudadano contribuyente madre padre conyuge conviviente contacto referencia victima denunciante procesado testigo medico docente socio heredero donante tutor jubilado pensionista derechohabiente customer patient employee member holder mother father spouse contact person student driver owner full complete last first
E4	ip mac email e_?mail correo electr[oó]nica web url tipo id general regional
E5	ip mac email e_?mail web url type id
E6	id cd cod codigo tipo fecha estado num nro
E7	id tipo codigo fecha estado cola log

APPENDIX D

Profile set and classifiers

The profile set `EC - LOPDP - v1` runs these 141 classifiers, with an assignment threshold of 60%.

Identificadores directos, cuasi-identificadores, datos sensibles (art. 4), datos de salud, datos crediticios y pasado judicial, conforme a la Ley Orgánica de Protección de Datos Personales del Ecuador y su reglamento.

CLASSIFIER	KIND	DOMAIN
EC_L1_CEDULA - Path	COLUMN NAME	EC_L1_CEDULA
EC_L1_CEDULA - Type	DATA TYPE	EC_L1_CEDULA
EC_L1_CEDULA - Regex	VALUE FORMAT	EC_L1_CEDULA
EC_L1_RUC - Path	COLUMN NAME	EC_L1_RUC
EC_L1_RUC - Regex	VALUE FORMAT	EC_L1_RUC
EC_L1_DOCUMENTO_EXTRANJERO - Path	COLUMN NAME	EC_L1_DOCUMENTO_EXTRANJERO
EC_L1_PASAPORTE - Path	COLUMN NAME	EC_L1_PASAPORTE
EC_L1_PASAPORTE - Regex	VALUE FORMAT	EC_L1_PASAPORTE
EC_L1_ACTA - Path	COLUMN NAME	EC_L1_ACTA
EC_L1_SEGURO_SOCIAL - Path	COLUMN NAME	EC_L1_SEGURO_SOCIAL
EC_L1_CODIGO_ESTUDIANTE - Path	COLUMN NAME	EC_L1_CODIGO_ESTUDIANTE
EC_L1_REGISTRO_PROFESIONAL - Path	COLUMN NAME	EC_L1_REGISTRO_PROFESIONAL
EC_L1_NOMBRE - Path	COLUMN NAME	EC_L1_NOMBRE
EC_L1_NOMBRE - List	KNOWN VALUES	EC_L1_NOMBRE
EC_L1_APELLIDO - Path	COLUMN NAME	EC_L1_APELLIDO
EC_L1_APELLIDO - List	KNOWN VALUES	EC_L1_APELLIDO
EC_L1_NOMBRE_COMPLETO - Path	COLUMN NAME	EC_L1_NOMBRE_COMPLETO
EC_L1_NOMBRE_COMPLETO - List	KNOWN VALUES	EC_L1_NOMBRE_COMPLETO
EC_L1_RAZON_SOCIAL - Path	COLUMN NAME	EC_L1_RAZON_SOCIAL
EC_L1_EMAIL - Path	COLUMN NAME	EC_L1_EMAIL
EC_L1_EMAIL - Type	DATA TYPE	EC_L1_EMAIL
EC_L1_EMAIL - Regex	VALUE FORMAT	EC_L1_EMAIL
EC_L1_TELEFONO - Path	COLUMN NAME	EC_L1_TELEFONO
EC_L1_TELEFONO - Regex	VALUE FORMAT	EC_L1_TELEFONO
EC_L1_DIRECCION - Path	COLUMN NAME	EC_L1_DIRECCION
EC_L1_DIRECCION - Type	DATA TYPE	EC_L1_DIRECCION
EC_L1_DIRECCION - Regex	VALUE FORMAT	EC_L1_DIRECCION
EC_L1_DIRECCION_COMPLEMENTO - Path	COLUMN NAME	EC_L1_DIRECCION_COMPLEMENTO
EC_L1_CUENTA_BANCARIA - Path	COLUMN NAME	EC_L1_CUENTA_BANCARIA
EC_L1_TARJETA - Path	COLUMN NAME	EC_L1_TARJETA
EC_L1_TARJETA - Regex	VALUE FORMAT	EC_L1_TARJETA

CLASSIFIER	KIND	DOMAIN
EC_L1_BILLETERA – Path	COLUMN NAME	EC_L1_BILLETERA
EC_L1_BILLETERA – Regex	VALUE FORMAT	EC_L1_BILLETERA
EC_L1_IP – Path	COLUMN NAME	EC_L1_IP
EC_L1_IP – Regex	VALUE FORMAT	EC_L1_IP
EC_L1_DISPPOSITIVO – Path	COLUMN NAME	EC_L1_DISPPOSITIVO
EC_L1_DISPPOSITIVO – Regex	VALUE FORMAT	EC_L1_DISPPOSITIVO
EC_L1_COOKIE – Path	COLUMN NAME	EC_L1_COOKIE
EC_L1_PLACA – Path	COLUMN NAME	EC_L1_PLACA
EC_L1_PLACA – Regex	VALUE FORMAT	EC_L1_PLACA
EC_L1_VEHICULO – Path	COLUMN NAME	EC_L1_VEHICULO
EC_L1_VEHICULO – Regex	VALUE FORMAT	EC_L1_VEHICULO
EC_L1_INMUEBLE – Path	COLUMN NAME	EC_L1_INMUEBLE
EC_L1_CONTRATO – Path	COLUMN NAME	EC_L1_CONTRATO
EC_L1_USUARIO – Path	COLUMN NAME	EC_L1_USUARIO
EC_L1_CREDENCIAL – Path	COLUMN NAME	EC_L1_CREDENCIAL
EC_L1_GEOLOCALIZACION – Path	COLUMN NAME	EC_L1_GEOLOCALIZACION
EC_L1_GEOLOCALIZACION – Regex	VALUE FORMAT	EC_L1_GEOLOCALIZACION
EC_L1_TEXTO_LIBRE – Path	COLUMN NAME	EC_L1_TEXTO_LIBRE
EC_L1_TEXTO_LIBRE – Type	DATA TYPE	EC_L1_TEXTO_LIBRE
EC_L1_TEXTO_LIBRE – Regex	VALUE FORMAT	EC_L1_TEXTO_LIBRE
EC_L2_FECHA_NACIMIENTO – Path	COLUMN NAME	EC_L2_FECHA_NACIMIENTO
EC_L2_FECHA_NACIMIENTO – Type	DATA TYPE	EC_L2_FECHA_NACIMIENTO
EC_L2_ANIO_NACIMIENTO – Path	COLUMN NAME	EC_L2_ANIO_NACIMIENTO
EC_L2_ANIO_NACIMIENTO – Type	DATA TYPE	EC_L2_ANIO_NACIMIENTO
EC_L2_EDAD – Path	COLUMN NAME	EC_L2_EDAD
EC_L2_EDAD – Type	DATA TYPE	EC_L2_EDAD
EC_L2_FECHA_EVENTO – Path	COLUMN NAME	EC_L2_FECHA_EVENTO
EC_L2_FECHA_EVENTO – Type	DATA TYPE	EC_L2_FECHA_EVENTO
EC_L2_SEXO – Path	COLUMN NAME	EC_L2_SEXO
EC_L2_SEXO – List	KNOWN VALUES	EC_L2_SEXO
EC_L2_PARROQUIA – Path	COLUMN NAME	EC_L2_PARROQUIA
EC_L2_PARROQUIA – List	KNOWN VALUES	EC_L2_PARROQUIA
EC_L2_DPA – Path	COLUMN NAME	EC_L2_DPA
EC_L2_DPA – Regex	VALUE FORMAT	EC_L2_DPA
EC_L2_CIUDADELA – Path	COLUMN NAME	EC_L2_CIUDADELA
EC_L2_CODIGO_POSTAL – Path	COLUMN NAME	EC_L2_CODIGO_POSTAL
EC_L2_CODIGO_POSTAL – Type	DATA TYPE	EC_L2_CODIGO_POSTAL

CLASSIFIER	KIND	DOMAIN
EC_L2_ESTADO_CIVIL – Path	COLUMN NAME	EC_L2_ESTADO_CIVIL
EC_L2_ESTADO_CIVIL – List	KNOWN VALUES	EC_L2_ESTADO_CIVIL
EC_L2_OCUPACION – Path	COLUMN NAME	EC_L2_OCUPACION
EC_L2_EMPLEADOR – Path	COLUMN NAME	EC_L2_EMPLEADOR
EC_L2_NIVEL_INSTRUCCION – Path	COLUMN NAME	EC_L2_NIVEL_INSTRUCCION
EC_L2_NIVEL_INSTRUCCION – List	KNOWN VALUES	EC_L2_NIVEL_INSTRUCCION
EC_L2_INSTITUCION_EDUCATIVA – Path	COLUMN NAME	EC_L2_INSTITUCION_EDUCATIVA
EC_L2_PERSONAS_A_CARGO – Path	COLUMN NAME	EC_L2_PERSONAS_A_CARGO
EC_L2_PERSONAS_A_CARGO – Type	DATA TYPE	EC_L2_PERSONAS_A_CARGO
EC_L3_ETNIA – Path	COLUMN NAME	EC_L3_ETNIA
EC_L3_ETNIA – List	KNOWN VALUES	EC_L3_ETNIA
EC_L3_IDENTIDAD_CULTURAL – Path	COLUMN NAME	EC_L3_IDENTIDAD_CULTURAL
EC_L3_IDENTIDAD_CULTURAL – List	KNOWN VALUES	EC_L3_IDENTIDAD_CULTURAL
EC_L3_LENGUA – Path	COLUMN NAME	EC_L3_LENGUA
EC_L3_LENGUA – List	KNOWN VALUES	EC_L3_LENGUA
EC_L3_CONDICION_MIGRATORIA – Path	COLUMN NAME	EC_L3_CONDICION_MIGRATORIA
EC_L3_CONDICION_MIGRATORIA – List	KNOWN VALUES	EC_L3_CONDICION_MIGRATORIA
EC_L3_NACIONALIDAD – Path	COLUMN NAME	EC_L3_NACIONALIDAD
EC_L3_NACIONALIDAD – List	KNOWN VALUES	EC_L3_NACIONALIDAD
EC_L3_RELIGION – Path	COLUMN NAME	EC_L3_RELIGION
EC_L3_RELIGION – List	KNOWN VALUES	EC_L3_RELIGION
EC_L3_IDEOLOGIA – Path	COLUMN NAME	EC_L3_IDEOLOGIA
EC_L3_IDEOLOGIA – List	KNOWN VALUES	EC_L3_IDEOLOGIA
EC_L3_FILIACION_POLITICA – Path	COLUMN NAME	EC_L3_FILIACION_POLITICA
EC_L3_FILIACION_POLITICA – List	KNOWN VALUES	EC_L3_FILIACION_POLITICA
EC_L3_AFILIACION_SINDICAL – Path	COLUMN NAME	EC_L3_AFILIACION_SINDICAL
EC_L3_AFILIACION_SINDICAL – List	KNOWN VALUES	EC_L3_AFILIACION_SINDICAL
EC_L3_BIOMETRICO – Path	COLUMN NAME	EC_L3_BIOMETRICO
EC_L3_BIOMETRICO – Type	DATA TYPE	EC_L3_BIOMETRICO
EC_L3_GENETICO – Path	COLUMN NAME	EC_L3_GENETICO
EC_L3_ORIENTACION_SEXUAL – Path	COLUMN NAME	EC_L3_ORIENTACION_SEXUAL
EC_L3_ORIENTACION_SEXUAL – List	KNOWN VALUES	EC_L3_ORIENTACION_SEXUAL
EC_L3_IDENTIDAD_GENERO – Path	COLUMN NAME	EC_L3_IDENTIDAD_GENERO
EC_L3_IDENTIDAD_GENERO – List	KNOWN VALUES	EC_L3_IDENTIDAD_GENERO
EC_L3_VICTIMA – Path	COLUMN NAME	EC_L3_VICTIMA
EC_SALUD_IDENTIFICADOR – Path	COLUMN NAME	EC_SALUD_IDENTIFICADOR
EC_SALUD_COBERTURA – Path	COLUMN NAME	EC_SALUD_COBERTURA

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EC_SALUD_COBERTURA – List	KNOWN VALUES	EC_SALUD_COBERTURA
EC_SALUD_DIAGNOSTICO – Path	COLUMN NAME	EC_SALUD_DIAGNOSTICO
EC_SALUD_DIAGNOSTICO – Regex	VALUE FORMAT	EC_SALUD_DIAGNOSTICO
EC_SALUD_PROCEDIMIENTO – Path	COLUMN NAME	EC_SALUD_PROCEDIMIENTO
EC_SALUD_MEDICAMENTO – Path	COLUMN NAME	EC_SALUD_MEDICAMENTO
EC_SALUD_MEDICAMENTO – List	KNOWN VALUES	EC_SALUD_MEDICAMENTO
EC_SALUD_TEXTO_CLINICO – Path	COLUMN NAME	EC_SALUD_TEXTO_CLINICO
EC_SALUD_TEXTO_CLINICO – Type	DATA TYPE	EC_SALUD_TEXTO_CLINICO
EC_SALUD_RESULTADO_EXAMEN – Path	COLUMN NAME	EC_SALUD_RESULTADO_EXAMEN
EC_SALUD_VIH – Path	COLUMN NAME	EC_SALUD_VIH
EC_SALUD_GRUPO_SANGUINEO – Path	COLUMN NAME	EC_SALUD_GRUPO_SANGUINEO
EC_SALUD_GRUPO_SANGUINEO – List	KNOWN VALUES	EC_SALUD_GRUPO_SANGUINEO
EC_SALUD_DISCAPACIDAD – Path	COLUMN NAME	EC_SALUD_DISCAPACIDAD
EC_SALUD_DISCAPACIDAD – List	KNOWN VALUES	EC_SALUD_DISCAPACIDAD
EC_SALUD_OCUPACIONAL – Path	COLUMN NAME	EC_SALUD_OCUPACIONAL
EC_SALUD_OCUPACIONAL – List	KNOWN VALUES	EC_SALUD_OCUPACIONAL
EC_SALUD_REPRODUCTIVA – Path	COLUMN NAME	EC_SALUD_REPRODUCTIVA
EC_SALUD_MENTAL – Path	COLUMN NAME	EC_SALUD_MENTAL
EC_FIN_HISTORIAL_CREDITICIO – Path	COLUMN NAME	EC_FIN_HISTORIAL_CREDITICIO
EC_FIN_DEUDA – Path	COLUMN NAME	EC_FIN_DEUDA
EC_FIN_DEUDA – Type	DATA TYPE	EC_FIN_DEUDA
EC_FIN_INGRESOS – Path	COLUMN NAME	EC_FIN_INGRESOS
EC_FIN_INGRESOS – Type	DATA TYPE	EC_FIN_INGRESOS
EC_FIN_PATRIMONIO – Path	COLUMN NAME	EC_FIN_PATRIMONIO
EC_FIN_PATRIMONIO – Type	DATA TYPE	EC_FIN_PATRIMONIO
EC_FIN_PENSION – Path	COLUMN NAME	EC_FIN_PENSION
EC_FIN_PENSION – Type	DATA TYPE	EC_FIN_PENSION
EC_FIN_PROGRAMA_SOCIAL – Path	COLUMN NAME	EC_FIN_PROGRAMA_SOCIAL
EC_FIN_PROGRAMA_SOCIAL – List	KNOWN VALUES	EC_FIN_PROGRAMA_SOCIAL
EC_FIN_REGISTRO_SOCIAL – Path	COLUMN NAME	EC_FIN_REGISTRO_SOCIAL
EC_FIN_REGISTRO_SOCIAL – List	KNOWN VALUES	EC_FIN_REGISTRO_SOCIAL
EC_FIN_VIVIENDA – Path	COLUMN NAME	EC_FIN_VIVIENDA
EC_FIN_VIVIENDA – List	KNOWN VALUES	EC_FIN_VIVIENDA
EC_PENAL_ANTECEDENTES – Path	COLUMN NAME	EC_PENAL_ANTECEDENTES
EC_PENAL_EXPEDIENTE – Path	COLUMN NAME	EC_PENAL_EXPEDIENTE
EC_PENAL_EXPEDIENTE – Regex	VALUE FORMAT	EC_PENAL_EXPEDIENTE